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Advancing adversarial and LLM robustness in trustworthy AI: a comprehensive survey

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Abstract
Despite significant advancements in various fields in recent years, artificial intelligence (AI) models have demonstrated strong performance and broad application potential. However, they still face numerous challenges in terms of security and robustness in practical applications. Among these, robustness stands out as a critical factor contributing to the perceived untrustworthiness of AI models and remains a major barrier to their widespread adoption. Moreover, most current AI models are designed with a black-box structure, lacking sufficient explainability, which makes it difficult for researchers to understand their decision-making mechanisms. This 'invisible' nature not only limits the ability to predict model behavior but also increases the instability of models in complex and unknown environments. In this paper, we systematically review the evaluation methods and enhancement strategies for AI model robustness from multiple perspectives: (1) We identify the primary issues and technical challenges in the robustness of current AI models. (2) We explore the connections and distinctions between core concepts of trustworthy AI. (3) We summarize the development of robustness evaluation in recent years from both the perspective of robustness evaluation metrics and methods. (4) We examine robustness enhancement methods across different stages of the AI model lifecycle: data preprocessing, training, model architecture design, and post-processing. (5) We focus on hallucinations and other robustness issues faced by generative large language models (LLMs), summarizing current research progress and mitigation strategies. (6) Finally, we discuss open questions and future research directions in the field of AI model robustness.

Keywords Trustworthy artificial intelligence · Robustness · Deep learning · Large language model

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1 Introduction

AI has emerged as one of the most groundbreaking technologies in recent years, with AI models making significant advancements in various fields (Tocchetti et al. 2025). For instance, automatic fire recognition assists in factory safety management; AlphaGo defeated the top human Go player (Angelov et al. 2021); AI models can drive our cars autonomously and even assist doctors and judges in making decisions about diseases or legal matters (Clement et al. 2023). In many areas, AI models have outperformed humans (Joshi et al. 2021). However, the rapid development and widespread application of AI have also brought about numerous challenges (Chander et al. 2025). One of the most pressing concerns is the security and robustness of AI models. The internal architecture of AI models remains a black box, lacking transparency, which makes it difficult for researchers to understand and interpret the models' decision-making and outputs (Vilone et al. 2020). For example, researchers can mislead image recognition models by using carefully designed adversarial samples to induce incorrect judgments (Chakraborty et al. 2018); LLMs may generate content that appears plausible but is factually inaccurate, known as the 'hallucination' phenomenon (Huang L et. 2025). Due to the lack of sufficient theoretical research and proof methods, AI models have yet to be widely applied, particularly in critical safety areas such as healthcare and finance. Therefore, it is necessary to try to understand the specific situation of how the AI model works inside, and understand the logic behind the algorithm and output results. Furthermore, ensuring the robustness of software systems that integrate AI, such as mobile applications is equally critical, Du et al. (2025) proposed a deep learning model for just-in-time defect localization that leverages pre-trained programming language encoders and hierarchical attention networks to precisely identify defective code lines.

AI models are highly susceptible to external attacks, leading to incorrect output results, which fails to meet our requirements for large-scale application of reliable technology (Speith et al. 2022). For instance, researchers required ChatGPT to repeatedly recite 'poems,' which led to the exposure of training data, including personal information such as phone numbers, emails, and actual addresses (Su et al. 2019); NVIDIA's AI team identified various vulnerabilities in machine learning code datasets, including insecure deserialization, XML injection, and improper handling of sensitive information (Kurakin et al. 2018); attacks on LLMs can involve creating specific prompts and using algorithms to align with the model's general and transferable adversarial attacks, leading to a new type of jailbreak backdoor attack, where the model accepts malicious or dangerous requests (Chen et al. 2022). To build truly trustworthy AI models, it is essential to ensure they adhere to principles of robustness, reliability, ethics, and fairness, ensuring correct output and predictions even under malicious attacks or extreme conditions. Therefore, exploring and addressing the robustness of AI models is a shared goal of both academia and industry. Currently, much research focuses on enhancing the adversarial robustness of models, which means improving their ability to resist external interference and attacks, with adversarial attacks and adversarial training being key areas of focus. However, these methods still fail to fundamentally understand the robustness issues of models. We found that most studies on robustness focus on four key areas: data preprocessing, optimization of training strategies, model architecture optimization, and post-processing. This paper refers to the methods used to enhance robustness during the data preprocessing stage as input data enhancement. Key factors contributing to reduced robustness include noise (Qian et al. 2019), non-adversarial perturbations (Gao et

al. 2020), false correlations (Wang et al. 2021), and distribution shifts (Zhang et al. 2022a, b). The primary method for data preprocessing is data transformation, which involves simple cropping and blurring (Mohammad et al. 2024). With the advent of generative models, many studies have started using diffusion models to generate new data, known as 'synthetic' data, which exhibits enhanced robustness (Islam et al. 2024; Wang et al. 2024a, b). Adversarial training is a primary method for optimizing training strategies, with techniques such as single-step iteration, multi-step iteration, and virtual adversarial training being continuously proposed and applied in practice. To enhance the robustness of unsupervised domain models, other authors (Zhang et al. 2023a, b, c, d) proposed a training algorithm based on the generative adversarial network (GAN): by alternately training the GAN model and the unsupervised model, it generates high-quality adversarial samples, effectively improving the model's performance under adversarial conditions. We categorize model architecture optimization and post-processing strategies as model enhancement. There are various methods for optimizing model architecture, and different modification strategies are applied to different original network structures, such as adding spatial or channel attention modules (Zhang et al. 2025). Common post-processing strategies include anomaly detection (Zhang et al. 2023a, b, c, d) and defense distillation (Papernot et al. 2016a, b). In addition, the robustness problem of LLMs cannot be ignored. Unlike traditional discriminative models whose robustness is often tested against input perturbations, due to their generative, autoregressive nature and training on next-token prediction, LLMs exhibit unique failure modes that strike at the core of their utility. Although there is no comprehensive method to enhance the robustness of LLMs, there have been quite a lot of research on hallucinations (Carlini et al. 2021), lack of context understanding (Agrawal et al. 2025), and lack of causal reasoning (Guo et al. 2024a, b). In the domain of automatic program repair, Liu et al. (2025) proposed ReAPR, a retrieval-augmented framework that mitigates hallucinations in LLM-generated fixes by incorporating relevant bug-fix pairs from a curated knowledge base, demonstrating the effectiveness of external knowledge integration in enhancing LLM robustness for specialized tasks.

This paper serves as a comprehensive roadmap for robustness in trustworthy AI. We navigate the vast literature to offer a structured analysis of how robustness can be assessed and fortified at every stage of an AI model's development. Crucially, it demystifies the distinct robustness profile of LLMs, integrating topics like hallucination mitigation into the core discussion. Through this synthesized and critical perspective, we aim to bridge the gap between isolated technical advances and the systemic design of reliable AI, concluding with a forward-looking analysis of the field's most pressing challenges and opportunities.

1.1 Comparing with existing surveys

Most existing papers focus on adversarial robustness, often overlooking systematic robustness and the connections and distinctions between robustness and other trustworthy AI attributes such as explainability and fairness. Tocchetti et al. (2025) focus on robust evaluation and robust reinforcement, but they do not explore robust techniques for generative LLMs. Angelov et al. (2021) provide a macro-level introduction to explainable AI, detailing the mathematical principles of AI security, but their scope is limited to fewer types of models, lacking research on the explainability of language models. Clement et al. (2023) offer a detailed introduction to explainable AI from a software application perspective, but the

review lacks explanations of relevant principles and robustness studies for LLMs. Joshi et al. (2021), while reviewing the explainability and robustness of AI models, do not provide a comprehensive overview of robust reinforcement methods. Bayram et al. (2025) comprehensively discusses methods to enhance adversarial robustness, detailing the latest adversarial attack and defense mathematical principles, and explores the relationship between model robustness and generalization performance, as well as the connection between adversarial training and data augmentation. However, this review lacks attention to non-adversarial robustness. In the other work (Jiang et al. 2025) focuses on the hallucination problem in LLMs, reviewing the causes and mitigation methods from multiple angles, but the authors do not explore other types of robustness issues in LLM. In summary, the current research on robustness is not sufficiently comprehensive compared to the rapidly advancing AI models, particularly the highly popular large language models of recent years. This review aims to address these shortcomings by exploring the relationship between trustworthy AI and robustness, providing a higher-dimensional and more comprehensive overview of the academic community's recent efforts in robustness evaluation and enhancement. The contribution of this paper lies not only in its broad coverage of robustness topics but also in its in-depth exploration of emerging areas, such as the robustness of generative LLMs. Table 1 provides a comparison between our review and existing reviews.

1.2 Systematic review methodology

To ensure a comprehensive and unbiased coverage of the article, this survey follows a systematic review methodology comprising three main stages: (1) search strategy formulation, (2) study selection based on predefined criteria, (3) quality assessment of included studies and classify each chapter. The process is visualized in Fig. 1.

Table 1 Comparison of other reviews. A comprehensive comparison showing our survey's coverage of trustworthy AI, evaluation methods, adversarial robustness, data enhancement, model enhancement, and LLM robustness relative to existing surveys

Survey	Research direction					
	Trustworthy AI	Evaluation	Adversarial	Data Enhance	Model Enhance	LLM
Our survey	√	√	√	√	√	√
Tocchetti et al. 2025	×	√	√	√	√	×
Clement et al. 2023	√	×	√	√	√	×
Wang et al. 2024	×	×	√	√	×	×
Mumuni et al. 2024	×	×	√	√	×	×
Kumar et al. 2024	×	√	√	√	×	×
Freitas et al. 2022	×	×	√	√	√	×
Goyal et al. 2023	×	×	√	√	√	×
Zühlke et al. 2025	×	√	√	√	√	×
Hung et al. 2024	×	×	√	√	√	×
Bayram et al. 2025	×	×	√	√	√	×
Dai et al. 2024	√	√	√	×	×	×
Jiang et al. 2025	×	√	√	√	×	√

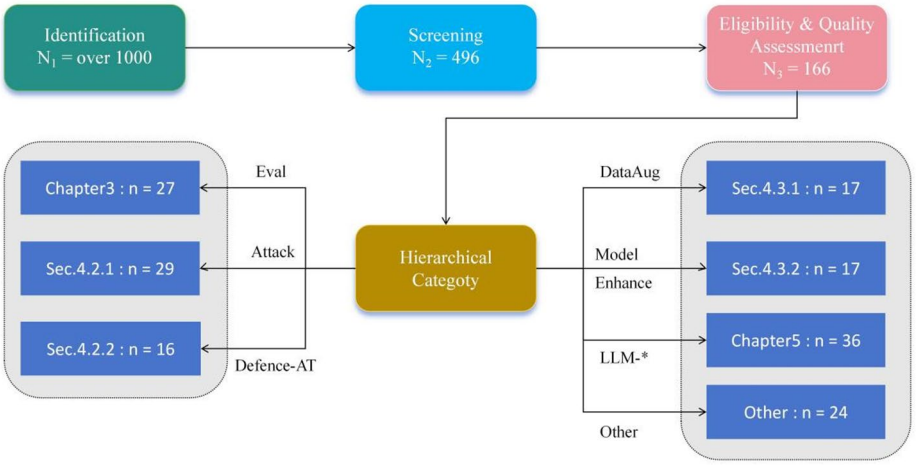


Fig. 1 Flowchart of the hierarchical systematic article review process. The process begins with identification and screening to build a master corpus, followed by a tagging and categorization step that allocates papers to specific technical sections of the survey based on their core contribution

1.2.1 Search strategy

To identify relevant studies, we conducted structured searches across five major academic databases: IEEE Xplore, ACM Digital Library, Web of Science, arXiv, and Google Scholar. The search timeframe was limited from January 2023 to May 2025 to focus on the most recent advancements. Meanwhile, we have also retained older but groundbreaking articles. The core search strings were constructed by combining terms from three conceptual groups using Boolean operators (AND/OR):

- Group 1 (core concept): "robustness" OR "adversarial robustness" OR "model robustness".
- Group 2 (AI domain): "artificial intelligence" OR "deep learning" OR "neural network" OR "large language model" OR "LLM".
- Group 3 (trustworthy AI attributes): "trustworthy AI" OR "explainability" OR "fairness" OR "generalization" OR "hallucination".

A representative search string used in IEEE Xplore is: ("adversarial robustness" OR "model robustness") AND ("deep learning" OR "large language model") AND ("trustworthy AI" OR "hallucination").

1.2.2 Study selection and eligibility criteria

The initial search yielded N_1 = over 1000 records. The selection process involved two sequential screening phases:

Phase 1: title and abstract screening. We screened titles and abstracts against the following inclusion criteria (IC) and exclusion criteria (EC):

- IC1: The study primarily focuses on evaluating or enhancing the robustness of AI models.
- IC2: The study is published in peer-reviewed conferences, journals, or reputable preprint servers (arXiv).
- IC3: The study is written in English.
- EC1: The study is a short paper (less than 4 pages), abstract only, tutorial, or survey (but we retained surveys for background and comparison in Sect. 1.1).
- EC2: The study's main contribution is not related to AI model robustness.
- EC3: The study is a duplicate publication.

Phase 2: full-Text review. The remaining papers underwent a full-text assessment against more stringent criteria:

- IC4: The study proposes a novel method, provides a significant empirical evaluation, or offers a clear theoretical analysis of robustness.
- EC4: The methodological description is insufficient to understand the robustness-related contribution.
- EC5: The empirical results are not reproducible or lack baseline comparisons.

After this two-phase screening, $N_2 = 496$ papers were selected for in-depth review and data extraction. The screening process and attrition reasons are detailed in the Fig. 1.

1.2.3 Quality assessment of included studies

To gauge the reliability and validity of the findings from the selected studies, we performed a quality assessment based on a set of questions adapted for AI robustness. Each study was evaluated on the following dimensions :

- **clarity of problem definition**: Does the paper clearly state the robustness problem it addresses?
- **methodological rigor**: Is the proposed method (attack, defense, evaluation) described in sufficient detail? Are the assumptions clear?
- **experimental evaluation**: Are the experiments comprehensive? Are standard benchmarks used? Are comparisons made against relevant state-of-the-art methods?
- **discussion of limitations**: Does the paper acknowledge its own limitations or potential negative societal impact?

Studies that scored highly across these dimensions (e.g., papers published in top-tier venues like CVPR, with open-sourced code and extensive ablation studies) were given more weight in our synthesis of evidence and identification of dominant trends. This assessment helped us critically interpret the results and avoid over-relying on findings from methodologically weak papers. In the end, we selected $N_3 = 166$ articles.

We assigned one or more technical tags to each article in the main library (N_3) and classified them into different technical chapters based on this. The specific focus process is as follows:

- First level classification (by model lifecycle and problem domain). We have preliminarily divided the article into three categories:

1. overview and conceptual papers (Chaps. 1 and 2): Used for background and framework construction.
2. general AI Model robustness (Chaps. 3 and 4): This mainly covers traditional DNN models such as vision and text.
3. robustness of large language models (Chap. 5): Mainly targeting Transformer based Generative LLM.

- Secondary classification (focus on technical chapters). Under the first level classification, more refined labeling is performed based on the core technical contributions of the article.

1. For the article on the robustness of general AI models, we use the following labels:

a. Evaluation indicators, evaluation methods → included in Chap. 3 (Robust Evaluation Techniques).
b. Attack white box, attack black box, Attack generative formula → classified under Sect. 4.2.1 (Adversarial Attacks). Selection focus: Priority should be given to pioneering methods such as fast gradient sign method (FGSM) (Goodfellow et al. 2014) and projected gradient descent (PGD) (Madry et al. 2017), as well as representative works in new paradigms such as attacks against vision transformers (ViTs) and physical world attacks.
c. Defense adversarial training, defense data enhancement, defense architecture optimization, and defense post processing are respectively classified in Sect. 4.2.2, 4.3.1, 4.3.2.1, and 4.3.2.2. Screening focus: Priority should be given to selecting methods that have reported significant gains on standard benchmarks (such as CIFAR-10-C) and have been widely cited in subsequent work.

2. For the article on "robustness of large language models", we use the following labels:

a. LLM illusion attribution, LLM illusion relief → classified in Sect. 5.1 (LLM Illusion Relief). Key screening points: covering the entire stage of mitigation strategies from data, training to inference, and ensuring that different solutions are included for factual hallucinations and fidelity hallucinations.
b. LLM robustness other (such as instruction robustness, multi agent, code security) → Classified in Sect. 5.2 (LLM Robustness Enhancement: Other Perspectives). Selection focus: Select research that can reflect the unique vulnerability of LLM and corresponding reinforcement schemes.

Through the above hierarchical classification process, the 162 articles included were systematically assigned to each chapter. Table 2 shows the distribution of article quantity for each technical sub topic, which intuitively reflects the research enthusiasm in different fields and ensures the balance of each part of the review.

Table 2 Distribution of article across survey chapters and sections

Chapter/ Section	Primary topic	Numbers of papers	Core tags
Chapter 3	Robustness Evaluation Techniques	27	[Evaluation - Metric], [Evaluation - Method]
Section 4.2.1	Adversarial Attacks	29	[Attack - White Box], [Attack - Black Box], [Attack - Generative]
Section 4.2.2	Adversarial Training	16	[Defense-AT]
Section 4.3.1	Input Data Enhancement	17	[Defense - Data Aug]
Section 4.3.2	Model Enhancement	17	[Defense - ArchOpt], [Defense - PostProc]
Chapter 5	Large Language Model Robustness	36	[LLM-*]
Other	Surveys, Concepts, Background	24	Framework and definition papers
Total		166	

1.3 Organization of this survey

In this paper, we first introduce the relevant concepts of AI robustness in Sect. 2 and systematically organize and summarize the relationships and differences among the core concepts of trustworthy AI. In Sect. 3, we introduce the testing indicators and evaluation methods for AI.

model robustness. By systematically summarizing these evaluation methods, we provide a foundational basis for analyzing the robustness defects of models in subsequent sections. After reviewing the evaluation indicators and methods for AI robustness, we gain a deeper understanding of the robustness issues present in the models through the evaluation results and address these issues with targeted enhancements. In Sect. 4, we comprehensively review the methods for enhancing AI robustness, starting from various stages of the model, and delve into common technical means such as input datasets augmentation, optimization of training strategies, model robustness. By systematically summarizing these evaluation methods, we provide a foundational basis for analyzing the robustness defects of models in subsequent sections. After reviewing the evaluation indicators and methods for AI robustness, we gain a deeper understanding of the robustness issues present in the models through the evaluation results and address these issues with targeted enhancements. In Sect. 4, we comprehensively review the methods for enhancing AI robustness, starting from various stages of the model, and delve into common technical means such as input datasets augmentation, optimization of training strategies, enhancement of model architecture design, and post-training processing. We provide a detailed analysis of their principles, advantages and disadvantages, and applicable scenarios. Furthermore, with the rapid development of generative LLMs, robustness issues have become increasingly prominent. So In Sect. 5, we focus on the robustness defects and enhancements of LLMs, gradually exploring new methods. Finally, we discuss several future research directions and challenges in robustness evaluation and enhancement, including but not limited to: how to improve existing evalu-

ation methods, how to achieve robust enhancement technology in complex scenarios, and cross-domain robustness research.

The contributions of this paper are as follows:

- A total of 166 representative papers on the robustness of AI in recent years have been collected and systematically sorted out, of which more than 80% are from 2023 to 2025, fully reflecting the latest research trends in this field.
- This paper summarizes the robustness evaluation and enhancement methods of AI models from multiple perspectives and stages, and constructs a structured research framework.
- The robustness of generative LLMs, especially hallucination phenomena, is systematically analyzed, and the current feasible mitigation strategies are summarized.
- The existing challenges in the field of robustness research are emphasized, and the prospective future research directions are proposed to provide theoretical support for subsequent work.

2 Related terms

trustworthy AI technology refers to AI that is correctly defined, legally compliant, and fair. We must ensure that the AI model can be safely described at every stage of its lifecycle, ensuring it is trustworthy and capable of reaching its full potential in terms of design, concept, intent, and outcome (Zhou et al. 2020). Robustness is a core concept in trustworthy AI, alongside transparency, privacy, fairness, generalization, and explainability. Figure 2 shows the six aspects of trustworthy AI.

2.1 Transparency

Transparency encompasses traceability, explainability, and communicability (Tocchetti et al. 2025). The data collection process, preprocessing, labeling, and application in decision-making should all be transparent (Hamidi et al. 2024).

2.2 Robustness

Robustness is one of the most critical concepts in trustworthy AI. It typically refers to a model's ability to withstand external attacks and environmental changes, which determines

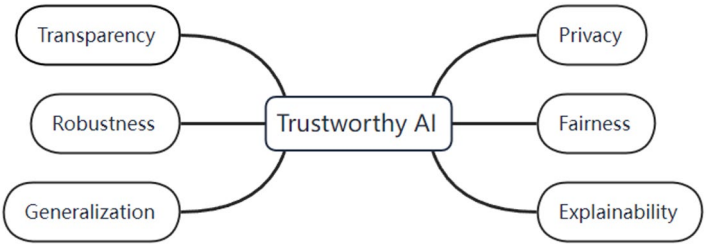


Fig. 2 Relevant concepts of trustworthy AI

its capability to handle errors, missing values, and incorrect data (Clement et al. 2023). In various subfields of AI, robustness has been discussed from different perspectives, leading to subtle differences in its definition. The fundamental definition of robustness is the model's performance being insensitive to parameter errors, meaning that the model's performance remains largely unchanged even when parameters are slightly altered. Currently, adversarial robustness and natural robustness are the two most discussed branches (Table 3).

Robustness against adversarial attacks refers to the model's ability to maintain its performance in the face of potential adversarial attacks or perturbations. Adversarial attack methods are diverse, primarily involving the input of maliciously perturbed data into AI models. These perturbations are imperceptible and non-random, aiming to mislead the model's decision-making process and maximize output errors. Various algorithms can be used for adversarial attacks, and the resulting samples are known as adversarial samples. These samples are created by using inputs that closely match normal samples in terms of certain metrics or similarities, leading to different model outputs (Wei et al. 2025; Freitas et al. 2022). Adversarial attacks can be categorized into targeted and non-targeted attacks, as well as black-box and white-box attacks, depending on the attacker's level of knowledge about the model. When a model is subjected to adversarial attacks, using adversarial samples for additional training can develop effective defense mechanisms, known as adversarial training. This training optimization strategy (Devaguptapu et al. 2021; Waghela et al. 2024) is widely applied in the field of model defense.

An example of the attack is shown in Fig. 3. By adding specific adversarial perturbations to the original "deer" image, the image recognition model produces a wrong classification of "horse".

Natural robustness refers to the model's ability to maintain its performance under natural conditions. Natural disturbances, such as changes in light intensity or visibility, introduce common types of natural noise into the input data, which are more likely to occur in real-world scenarios. The corresponding defense measures are similar to those for adversarial robustness.Local and global robustness are also key areas of focus in many studies. Specifically, local robustness is defined within the neighborhood of a sample, referring to the range of variations and perturbations around it. Zhang et al. (2020) introduced the concept of local adversarial robustness in machine learning systems, where it is defined as a unique scenario that explains how perturbations are generated or their origins. Global robustness lacks a consistent definition, and there is currently a significant lack of methods for analyzing, evaluating, and quantifying global robustness. As discussed (Zhang et al. 2023a, b, c,

Table 3 Core concepts of trustworthy AI and their relationships with robustness

Trustworthy AI	Concept	Correlation with robustness
Transparency	Training data is traceable and visible	Positive
Robustness	Resist attacks and environmental changes	Positive
Generalization	Make accurate with unseen data	Uncertain
Privacy	User privacy, data quality, and data integrity	Uncertain
Fairness	Avoid bias, accessibility, and universal design	Uncertain
Explainability	Human understandable information	Positive

The table defines six key concepts (transparency, robustness, generalization, privacy, fairness, explainability) and indicates their correlation (positive or uncertain) with robustness



Fig. 3 Example of an adversarial attack on image classification. Note: The adversarial perturbation is magnified for visual clarity."

d) about the definition of global robustness applies to all inputs and can be measured by the statistical representation of the local robustness radius, which describes the overall robustness level of the model.

2.3 Generalization

Generalization ability is closely linked to the concept of robustness. It refers to the model's capability to make accurate predictions or decisions when faced with unseen data, which is a key metric for evaluating model performance. Both robustness and generalization ability reflect the reliability of models in practical applications; enhancing generalization ability typically also indirectly improves robustness (Xiao et al. 2018).

2.4 Privacy

Privacy in the field of AI encompasses respecting user privacy, ensuring data quality and integrity, and securing access to data (Sarkar et al. 2017). There is a complex dual relationship between privacy and robustness: on the one hand, technologies such as differential privacy can enhance robustness; On the other hand, excessive privacy protection may limit the model's ability to adapt to abnormal inputs.

2.5 Fairness

Fairness is a crucial standard for trustworthy AI, encompassing the avoidance of bias, accessibility, and universal design. AI-designed models or systems must be fair and neutral, with training datasets free from bias, ensuring they can make equitable decisions after appropriate training. If the training data contains real-world biases, the model must learn, detect, and address these biases on a large scale (Hinton et al. 2015). Researchers in this field have developed several fairness metrics and methods to mitigate unfairness. Moreover, there is a relationship between robustness and fairness: the impact of fairness metrics and methods on the robustness to different types of adversarial and natural perturbations; robustness enhancement methods also influence fairness.

2.6 Explainability

Explainability in the field of AI is defined as the ability to transform implicit information into human-understandable information. A trustworthy AI model should be comprehensible to humans. For simple models like linear regression or decision trees, they are inherently intuitive and easy to understand (Wu et al. 2024). By analyzing the decision-making process of the model, potential weaknesses or vulnerabilities can be more easily identified, allowing for measures to improve the model's robustness (Yang et al. 2024). However, it is important to note that increasing a model's explainability does not always enhance its robustness, and enhancing robustness does not necessarily improve a model's explainability.

In this section, we discuss the relevant concepts of trustworthy AI related to robustness and the relationship between each concept and robustness. In the next section, we will review the evaluation indicators and methods for AI model robustness.

3 Robustness evaluation techniques

3.1 Robustness evaluation index

The evaluation of the robustness of an AI model depends on various indicators, which depend on the evaluation method adopted and provide support for the robustness of the model from different perspectives. However, different types of model architectures and different tasks require different evaluation indicators.

The robustness metrics provided by the robustness benchmark evaluation method are based on the correctness of the benchmark. For most tasks, common evaluation metrics include confusion matrices, accuracy, precision (Westgard et al. 1981), recall, F-measure (Yacouby et al. 2020), ROC curves, and AUC curves, among others (Mao et al. 2019). In object detection tasks, the intersection-over-union (IoU) is often used to measure the prediction overlap of target detection boxes, while the mean average precision (mAP) is one of the most commonly used performance metrics in object detection tasks. It represents the average of the mean precision across multiple categories or detection thresholds, taking into account changes in both precision and recall (Buckland et al. 1994). In semantic segmentation tasks, common evaluation metrics include pixel accuracy and mean intersection-overlap (Poursaeed et al. (2018). In object tracking tasks, common evaluation metrics include center error, overlap rate, and success rate. In machine translation tasks, the bilingual evaluation understudy (BLEU) score is used to compare the similarity between machine translation results and reference texts (Lu et al. 2020). Basic performance indicators provide benchmarks for evaluation. As shown in Table 4, different tasks have their own standard metrics. The definitions and calculations of these indicators are widely known and there are standard implementations.

In addition to these classic robustness evaluation metrics, the robustness boundary is also a local robustness measure derived from neural network verification. It is defined as the verified hypersphere within the input space, where the maximum safe radius represents the robustness limit (Zhang et al. 2023a, b, c, d) for a given input point. Typically, this boundary's radius is used to quantify the level of robustness. When both exact and approximate verification methods are available, the exact boundary and the approximate lower bound of

Table 4 Common evaluation metrics for different AI tasks

Type of task	Common evaluation indicators	Core focus
Image classification	Accuracy, recall and F1 scores	reduce FP、FN
Object detection	The intersection ratio is compared with IoU, precision AP and average precision mAP	Sorting quality and category coverage
Semantic segmentation	Compare the intersection to the IoU	Degree of border overlap
Example segmentation	Accuracy, AP	Sorting quality and category coverage
Object tracking	Central point error (Yang et al. 2017)	Positional accuracy
Text categorization	Accuracy, recall and F1 score	Reduce FP、FN
Machine translation	BLEU fraction (Chang et al. 2025)	N-gram matching and length penalty
Speech recognition	Word error rate	Reduce error rate
Image production	Fréchet inception distance (FID) (Jayasumana et al. 2024)	Feature distribution similarity
Reinforcement learning	Cumulative rewards, convergence speed	Long-term return maximization, training efficiency

For each task type (image classification, object detection, semantic segmentation, etc.), the table lists the most relevant evaluation indicators and their core focus

the robustness boundary can be determined separately. Furthermore, adversarial testing is used to calculate the exact lower bound of input domain perturbations that lead to incorrect model outputs, such as pointwise robustness (Fakorede et al. 2023). Thus, the radius of the local robustness upper bound can be represented by the minimum adversarial distance of this exact boundary (Pang et al. 2020). There are two types of global robustness measures: those based on local robustness statistics and those based on benchmark correctness. Global robustness can be estimated through the statistical analysis of local robustness metrics from multiple test input points. The most commonly used local robustness metric is the L_p norm, particularly the radius of the hypersphere. Therefore, global robustness is also measured using these radius statistics, with existing measures including mean, minimum, expectation, and threshold-based statistics (Liang et al. 2022). Measures based on benchmark test correctness provide robustness metrics derived from robustness benchmark test evaluation methods.

The L_p norm, including L_0, L_1, and L_∞, uses distance metrics to assess the extent of input variation introduced by attacks. These norms measure the difference (Bayer et al. 2023) between the original input and the adversarial examples used to deceive the model. The L_0 norm indicates the number of pixel values or feature changes in the adversarial attack setting. Minimizing the L_0, norm aims to minimize the modification of the input. However, due to its non-differentiability, optimizing this norm can be challenging (Reyes-Amezcua et al. 2024). The L_1 norm calculates the sum of the absolute differences between each perturbation component and its corresponding initial input element. Minimizing the L_1 norm promotes sparse perturbations, where only a few features undergo significant

changes (Chen et al. 2024). The L_2 norm calculates the square root of the sum of the squared differences between each perturbation element and its corresponding element in the original input. This measures the Euclidean distance between the perturbation and the original input. The L_∞ norm calculates the maximum absolute difference between any two corresponding elements in the perturbation and the input. Minimizing the L_∞ norm promotes perturbations that modify only a few features while still allowing for significant changes (Allen-Zhu Z and Li. 2022).

Beyond the task-specific metrics summarized above, researchers have also proposed general-purpose benchmarks and diagnostic tools for holistic robustness assessment. Hendrycks and Dietterich (2019) established a seminal benchmark for evaluating model robustness to common natural corruptions and perturbations (noise, blur, weather effects), providing a standardized suite to measure "natural robustness." They proposed two complementary benchmarks—ImageNet-C for common corruptions and ImageNet-P for common perturbations—each with standardized metrics that shift the focus from worst-case adversarial robustness to average-case performance degradation. The key metrics of ImageNet-C are:

- Corruption error (CE): For a given corruption type cc, it is the sum of a model's top-1 error rates across five severity levels, normalized by the performance of a baseline model on the same corruption:

$$CE_c^f = \frac{\sum_{s=1}^{5} E_{s,c}^f}{\sum_{s=1}^{5} E_{s,c}^{AlexNet}} \tag{1}$$

  A lower CE indicates better robustness to that specific corruption.
- Mean corruption error (mCE): The primary aggregate metric, computed as the average of the CE scores across all 15 corruption types. It provides a single scalar measure of overall corruption robustness.
- Relative corruption error & relative mCE: These variants subtract the model's clean data error rate before normalization. The Relative mCE isolates the performance degradation due to the corruptions themselves, revealing that while modern architectures have improved in absolute mCE, their relative robustness (i.e., the drop in accuracy when corrupted) has seen negligible gains from AlexNet to ResNet.
  ImageNet-P consists of sequential perturbations (e.g., gradual rotation, zoom, brightness change) applied to images to test prediction stability. Its core metrics evaluate consistency rather than absolute accuracy:
- Flip probability (FP)/Flip rate (FR): The FP measures the probability that a model's top-1 prediction changes between two consecutively perturbed frames in a sequence. The FR standardizes this rate by the FP of AlexNet:

$$FR_p^f = \frac{FP_p^f}{FP_p^{AlexNet}} \tag{2}$$

- Mean flip rate (mFR): The average of FR values across different perturbation types, quantifying overall top-1 prediction instability.

- Top-5 distance (TSD) & mean TSD (mTSD): These metrics quantify changes in the model's top-5 prediction ranking across perturbations. A distance function calculates the displacement of class rankings between frames. The mTSD averages this across perturbations, capturing subtler forms of output inconsistency beyond the top-1 label flip.

Complementing this, Santos et al. (2024) introduced a signal-to-noise metric that quantifies the relative importance of foreground versus background features in adversarially trained models, thereby linking robustness evaluation with model interpretability. These works expand the toolbox for robustness evaluation beyond adversarial settings.Their evaluation framework is realized through the following metrics:

- Attribution map consistency: The study quantifies the similarity of decision rationales across models by calculating the Spearman's rank correlation between saliency maps generated for the same input by different models. A key empirical finding is that models trained adversarially exhibit significantly higher correlation in their attribution maps with each other—even across different architectures—compared to standardly trained models. This suggests that adversarial training drives models towards more consistent and interpretable feature representations, where consistency may signal learning of more essential features.
- Foreground-background signal-to-noise ratio: To directly measure whether a model focuses on semantically relevant regions, the authors introduce an Attribution Signal-to-Noise Ratio (in dB). This metric involves manually segmenting the foreground (target object) and background of an image, computing the mean attribution score for each region, and defining SNR as:

$$SNR\,(dB) = 10 \cdot log_{10}\left(\frac{\mu_F}{\mu_B}\right) \tag{3}$$

A higher SNR indicates greater focus on the target object versus spurious background correlations. The research found that adversarial training significantly increases SNR, and for certain attribution methods, SNR increases with model prediction confidence, providing empirical evidence for "confidence for the right reasons." This metric successfully links robustness assessment directly to interpretability quality.

3.2 Adversarial robustness evaluation methods

This section reviews evaluation methods specifically designed for adversarial robustness. Early adversarial robustness evaluation methods for AI models aimed to obtain precise solutions of the robustness boundary. These methods typically employed techniques such as satisfiability modulo theory (SMT) or Mixed Integer Programming to rigorously assess the robustness of AI models against all possible inputs, known as completeness evaluation. To address the computational complexity of completeness evaluation, researchers introduced incomplete evaluation methods (Li et al. 2023a, b, c). These methods aim to calculate a lower bound that approximates the precise boundary, ensuring consistent outputs for inputs within the verified region. The convex relaxation approach uses simpler convex functions to approximate the behavior of neural networks, simplifying the proof process. These meth-

ods transform the problem of solving the robustness boundary into a convex optimization problem (Dvijotham et al. 2018). This method often employs techniques like semidefinite programming (SDP) (Raghunathan et al. 2018).

Tsui-Wei Weng et al. (Jain et al. 2020) introduced a novel robustness evaluation metric, CLEVER, which is a computationally feasible and attack-independent algorithm for large neural networks, supported by the application of extremum theory. Chang et al. (2020) tested and evaluated the metric on three different datasets and models, addressing the bias issue in adversarial attacks. They calculated robustness by measuring the correctness difference between the original AI model and the perturbed AI model. In the same line, Guo et al. (2025) introduced a new robustness evaluation metric, for deep learning models under adversarial attacks. This metric considers the adversarial super volume, which comprehensively reflects the model's overall resistance to interference, unlike traditional methods that only consider model performance at specific perturbation intensities. Additionally, the authors proposed a new adversarial super volume attack and training method, first identifying a series of instances with the lowest confidence scores to describe the adversarial frontier. Then, the volume of the frontier is calculated using known algorithms from multi-objective optimization. The adversarial super-volume training adds the adversarial frontier examples found to the training dataset and gradually increases the perturbation level in conjunction with the adversarial super-volume evaluation framework. The Lipschitz constant (Zühlke et al. 2025; Ni et al. 2024) is estimated using a method based on the Lipschitz constant to quantify the model's sensitivity to small changes in input data, thereby estimating the lower bound of robust behavior. A smaller Lipschitz constant indicates that the model is less affected by input variations, suggesting higher robustness. This method is applicable to various network architectures, but one drawback is that the Lipschitz constant-based approach often results in a broader lower bound, which is far from the exact solution. The method based on random smoothing analyzes the robustness (Goli and Farnia, 2024) by introducing random noise into the input to observe its impact on the output. Monte Carlo sampling is used to generate samples around the input data, and a method for ensuring top-1 prediction robustness is proposed. The authors enhance image visibility by calculating the average value of pixels within each segmentation group using random smoothing based on pixel segmentation.

Gala et al. (2024) reveals that traditional robustness evaluation methods often lack scalability. The authors propose a method to measure adversarial subspaces, starting with three assumptions: the size of adversarial subspaces is small and limited; each test point has multiple directly related adversarial subspaces; and adversarial subspaces are self-contained and fully enclosed by the input space. This research addresses the challenge in traditional robustness evaluations where different modalities cannot be transferred.In Some adversarial training methods aim to improve multiple facets of robustness simultaneously; for instance, Santos et al. (2023) proposed an approach to enhance both background/foreground color robustness and model interpretability.

The robustness evaluation of LLMs focuses on their adaptability to complex tasks. Yu et al. (2025) developed a framework for generating multi-task test data, including five roles and their interaction mechanisms, as well as how to use this framework to generate data for single and multiple tasks. The authors proposed a dynamic evaluation method that manually identifies tool dependencies and uses a depth-first search algorithm to generate all possible execution paths, thereby verifying the correctness and optimality of the large language

model. Levy et al. (2025) also addressed the vulnerability of large predictive models to adversarial attacks, proposing a framework called robustness measurement and assessment (RoMA) to quantify the resistance of LLM to adversarial inputs. This framework implements a semantic perturbation evaluation method based on word embeddings, which generates sentences with semantic changes by replacing words in the input sentence. These sentences are then passed to a pre-trained LLM to obtain prediction confidence scores. The second highest confidence score distribution is analyzed to determine if it follows a normal distribution, and Box-Cox power transformation is applied to normalize the data for reliable statistical analysis.

This chapter systematically reviews the evaluation indicators and methods for the robustness of AI models. However, the current ecological assessment still faces several core challenges and trade-offs. Firstly, at the indicator level, the pursuit of robust assurance with strong authentication (such as through formal verification) often comes at the cost of huge computational overhead and conservative boundaries, and there is an inherent tension between it and benchmark tests oriented towards average performance (such as mCE). Secondly, at the methodological level, adversarial assessment and natural disturbance assessment constitute two complementary but not yet unified paradigms; The former is sensitive to carefully crafted attacks, while the latter evaluates its generalization ability to common corrosion, both of which fail to fully capture the complexity of the real world. In addition, existing evaluations mostly focus on classification tasks and static images, and the transferability and effectiveness across tasks (such as generation and decision-making) and dynamic environments urgently need to be verified. In the future, building a unified, efficient, and multidimensional evaluation framework, and exploring the correlation between evaluation results and the intrinsic mechanisms of the model will be the key to driving robustness research from “measurement” to “understanding”.

4 Robustness enhancement technology

4.1 Attack and defense of AI model

Most research focuses on various stages of AI models, exploring methods to enhance their robustness throughout the model’s lifecycle. Typically, the AI model lifecycle is divided into five key stages: data collection, data preprocessing, model training, model inference, and post-processing (Zhang et al. 2020), with an additional stage—fine-tuning—often included for LLMs (Gekhman et al. 2024). At each of these stages, the model may be vulnerable to different types of security threats, which are primarily categorized based on the attacker’s objectives and the impact of the attack into integrity breaches and availability breaches, where integrity breaches involve attackers manipulating the learning process to evade detection while allowing the model to function normally, making the compromise subtle and difficult to detect, and availability breaches aim to directly degrade or disrupt the model’s functionality, such as by poisoning the training data to reduce the performance of a target detection model, highlighting the importance of understanding these threats across the entire lifecycle in order to develop more secure and resilient AI systems.

The aim of an availability attack is to compromise the model, with attackers aiming to inflict maximum performance loss on the model at minimal cost (Kuang et al. 2024).

Table 5 Taxonomy of attacks on AI models based on attack goals

Attacks are categorized by their intended outcome (integrity violation vs. availability violation) and targeting strategy (targeted vs. any-target)

Type of attack		Target	
		Target attacks	Attack any target
The attack results	Output specific error results	Integrity violation	Availability violation
	Output any error results	Availability violation	Availability violation

Table 6 Attack vectors and defense strategies across the AI model lifecycle

For each stage (data collection, preprocessing, training, inference, post-processing), the table summarizes typical attack methods and corresponding defensive approaches

life cycle	The means of attack	Defense strategy
Data collection	Data distribution bias and fabricated data	Data detection, data authentication
Data preprocessing	Data poisoning	Data enhancement, feature compression
Model training	Backdoor attack, poison attack	Regularization, normalization
Model inference	Resist attacks on data integrity	Fight against purification, fight against training
Reprocessing	Model theft attack query attack	Anomaly detection, integrated learning

The attackers aim to create poisoned data that, when introduced during the training phase, results in model parameters that cause the model to perform poorly when tested without the poisoned data.

The purpose of an integrity violation attacker is to hide its attack behavior while making the model's output incorrect or achieving the attacker's goal (Allen-Zhu et al. 2022). Compared with a availability violation attacker, the optimization function of an integrity violation attacker differs in that the former requires the system to minimize the loss function except for the attacked data.

As summarized in Table 5, attacks on AI models can be fundamentally categorized by the attacker's primary objective. Integrity violations aim to cause specific, often hidden, misbehaviors while the model appears normal, whereas availability violations seek to generally degrade or deny the model's service. This taxonomy helps in understanding the threat model and devising appropriate stage-specific defenses, as discussed in the following lifecycle analysis.

In response to different attack methods, researchers have found different defense strategies. As shown in Table 6, common attack and defense methods at different stages of the AI model life cycle are listed.

The training effectiveness and performance of AI models largely depend on the quality and volume of the data they use, making the data collection and preprocessing stages critical yet vulnerable to various security threats (Jia et al. 2024). A common threat during these stages is the contamination or theft of training data by malicious actors. Among these, data poisoning stands out as a prevalent attack method, where attackers deliberately contaminate the training dataset to either degrade overall model performance or induce specific targeted errors (Tramèr et al. 2017). In the domain of image recognition, for instance, label flipping attacks manipulate the labels of training samples, thereby undermining the model's clas-

sification accuracy. Similarly, attackers may introduce hidden perturbations into the input data while keeping the labels unchanged, effectively altering the feature space in a way that is difficult to detect (Laugros et al. 2020). Another related technique is adversarial sample injection, which involves embedding adversarial examples—crafted using specific algorithms—into the training set with the intent of reducing the model's robustness. To counter such threats, defense strategies primarily focus on verifying the credibility of data sources and detecting or filtering out malicious or anomalous data, including approaches such as employing K-NN algorithms to identify and remove poisoned samples or conducting feature analysis to detect and eliminate abnormal data patterns.

During the training and inference phases of AI models, adversarial attacks and boundary attacks are among the most commonly employed methods. These attacks aim to misled the model by identifying and exploiting the smallest perturbations near the decision boundary, where minor input changes can lead to significant shifts in output (Zhang et al. 2024). By introducing carefully crafted noise that is often imperceptible to humans, attackers manipulate the model's predictions while remaining difficult to detect. These types of attacks have been extensively studied across various domains, including image recognition, natural language processing, speech recognition, and video analysis. Researchers (Chakraborty et al. 2018) have developed a wide range of attack algorithms tailored to different threat scenarios—for example, gradient-based methods such as FGSM and PGD are widely used in white-box settings, while techniques like single-pixel attacks and transfer-based attacks are effective in black-box environments. In response, numerous defense strategies have been proposed, including adversarial training, adversarial normalization, feature detection mechanisms, regularization techniques, and input normalization methods—all aimed at improving model robustness and mitigating the impact of such attacks.

After the model training is complete, the parameters can be compressed by identifying and removing unnecessary or unstable attributes of the model. One work developed a method to selectively replace ReLU neurons that are identified as unstable and useless (i.e., operating in the functional edge region) with linear activation functions, which helps enhance robustness at minimal performance cost (Papernot et al. 2016a, b). Ensemble learning and model fusion are also common defense strategies, where additional models are integrated into the already trained models to improve overall robustness through mutual enhancement.

We start by examining the robustness of adversarial methods, systematically discussing adversarial sample generation algorithms and defense strategies at the data level, and comparing various types of adversarial training strategies. We then introduce general data augmentation and model enhancement techniques, which are highly adaptable and can be applied to most robust reinforcement scenarios. Finally, we will explore several different post-processing robust reinforcement methods for models. Figure 4 clearly illustrates multiple aspects of the robustness enhancement strategy proposed in this article.

4.2 Adversarial robustness

Adversarial robustness refers to the ability of an AI model to withstand deliberate attacks or to deceive its prediction results, also known as adversarial example robustness. The attacks that challenge adversarial robustness are called adversarial attacks, which involve adding subtle, imperceptible perturbations to the input data, causing the AI model to produce

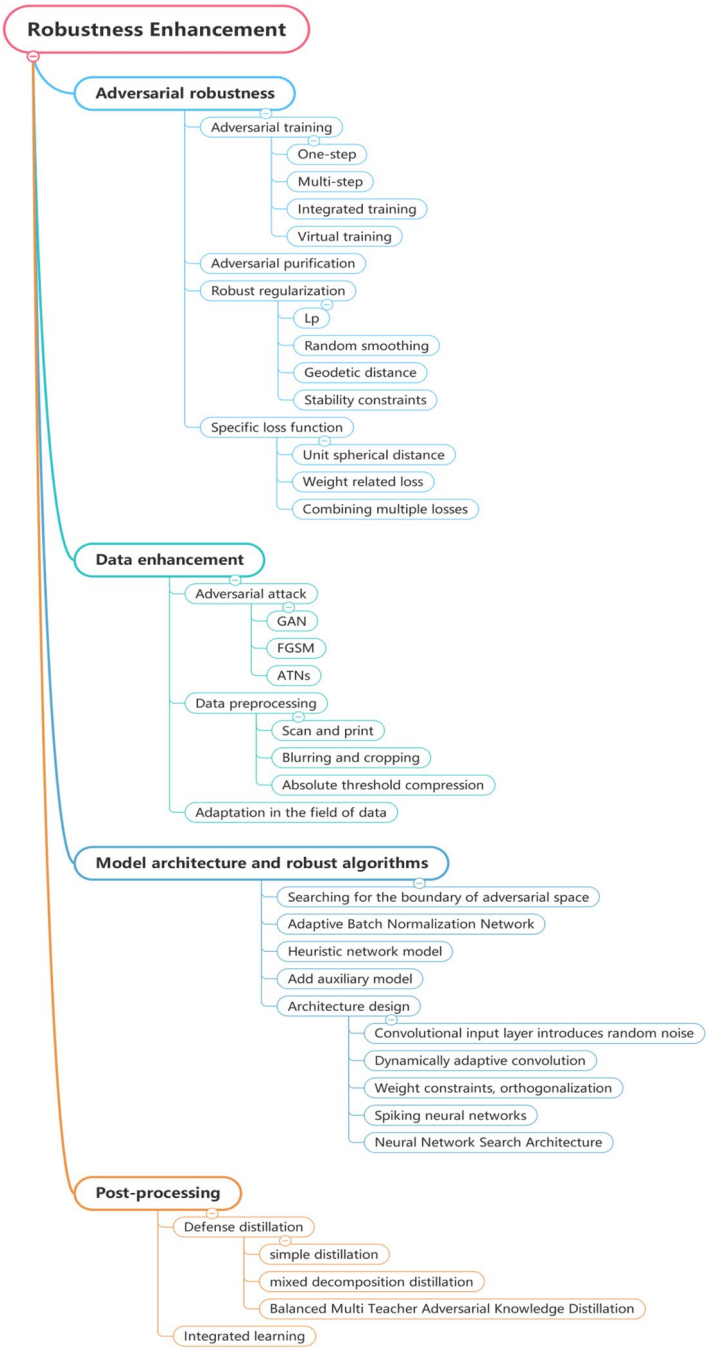


Fig. 4 Robust enhancement strategy. The framework categorizes techniques by their application stage in the AI model lifecycle

incorrect predictions or decisions. Therefore, adversarial robustness is a critical metric for assessing whether a model can accurately perform its intended tasks in such adversarial environments.

4.2.1 Adversarial attack

Adversarial attack is to attack AI models by generating adversarial examples, which is a data-enhanced robustness method (Xiao et al. 2018). According to the characteristics and effects of attacks, common adversarial attacks can be divided into white box attack and black box attack. Attacks can be categorized into single-step attacks, iterative attacks, targeted attacks, and non-targeted attacks, as well as specific perturbations and general perturbations (Baluja et al. 2017). In a white-box attack, the attacker has full access to the target model's structure and parameters, including the training data, gradient information, and activation functions. In a black-box attack, the attacker cannot access the model's structure or parameters and can only obtain the model's output by inputting raw data, from which they can infer the model's defense mechanisms and generate adversarial attacks. A single-step attack involves generating adversarial samples with a high success rate in a single computation (Zhou et al. 2020). An iterative attack requires multiple runs of the adversarial attack algorithm to produce adversarial examples. Compared to one-time attacks, iterative attacks take longer but yield better results. A targeted attack involves designing adversarial samples that cause the target classifier to misclassify a specific category, such as misclassifying category 7 as category 7. A non-targeted attack, also known as a non-targeted attack, involves the adversarial sample causing the classifier to misclassify any category other than the intended one, such as misclassifying category 7 as any incorrect category. Specific perturbations involve adding different perturbations to each input's original data, resulting in various perturbation patterns. General perturbations involve adding the same perturbation to each input's original data, resulting in identical perturbation patterns.

Early research on adversarial attacks revealed the vulnerability of deep neural networks to specific perturbations. In exploring the explainability of deep learning, the author discovered that the model's vulnerability to specific perturbations can lead to significant deviations in recognition results (Szegedy et al. 2013). This led to the development of the first adversarial attack algorithm for deep learning, L-BFGS. Thanks to its transferability, L-BFGS can be applied to models trained on different datasets, sparking a surge in research into adversarial examples among scholars. The FGSM algorithm was introduced to demonstrate that adversarial samples are generated by the high-dimensional linear nature of deep neural networks (Goodfellow et al. 2014). One of the key advantages of the FGSM algorithm is its fast attack speed, as it operates on a single-step basis. However, the success rate of adversarial samples generated through a single-step attack can be relatively low. To address this issue, Kurakin et al. proposed an iterative version of the FGSM (I-FGSM). The primary innovation of I-FGSM is the use of multiple small steps to incrementally increase the loss function, thereby generating more adversarial samples.

As research deepens, more attack methods have been proposed to meet the needs of different scenarios. Papernot et al. (2016a, b) introduced the jacobian-based significant map attack (JSMA). JSMA does not rely on the gradient information from the loss function of the model output but instead uses the probability information of the model's output categories for back propagation to obtain gradient information. In the line, a method was proposed

defensive distillation (Papernot et al. 2016a, b), which uses the distillation algorithm to transfer knowledge from a complex network to a simpler one, preventing attackers from directly targeting the original model. Defensive distillation effectively defended against some adversarial examples. In response to defensive distillation, Carlini et al. (2017) proposed the Carlini & Wagner (C&W) attack using L_0, L_2, and L_∞ constraints. The author of this work introduced a single-pixel attack (OPA) (Su et al. 2019), which can produce adversarial examples by changing just one pixel. This method uses a differential evolution algorithm to find adversarial perturbations, leading to model misclassification by altering one or a few pixels. Moosavi-Dezfooli et al. (2016) proposed a DeepFool method based on gradient iteration to generate adversarial perturbations. Compared to the single-step attack of FGSM, DeepFool can produce more accurate perturbations in a shorter time, but it only supports non-targeted attacks. Unlike some existing black-box attack methods based on proxy models, Chen et al. (2017) proposed Zeroth-order Optimization (ZOO), which does not leverage the attack transferability of proxy models. Instead, it estimates the values of first and second gradients and then uses Adma or Newton methods to iteratively find the best adversarial sample, adding perturbations to the given input. In the other work, Moosavi-Dezfooli et al. (2017) proposed a universal adversarial perturbation attack (UAP) based on DeepFool, which uses adversarial perturbations to push normal samples to the decision boundary, forming adversarial samples. The author of this work introduced an adversarial attack method called advGAN (Xiao et al. 2018), which is based on a GAN. In this line, Baluja et al. (2017) proposed an adversarial transformation network (ATN) based on a generative model. The ATN takes the original example as input, and the adversarial example is generated through a trained feedforward neural network. This approach ensures that the perturbations are minimal while maintaining the similarity between the adversarial and original examples, and also meets the success rate of adversarial attacks. Alone the same line,

Sarkar et al. (2017) introduced two black-box attack methods: UPSET and ANGRI. Cisse et al. (2017) proposed the Houdini algorithm, which can generate adversarial examples for the final performance measurement of tasks performed by the model, addressing issues such as speech recognition and semantic segmentation, where adversarial examples cannot be generated through gradient descent. In real-world tasks, pre-trained models are hard to obtain. One work proposed the data-free alternative training (DaST) method, which can generate adversarial black-box attack models without any real data (Athalye et al. 2018). To achieve this, DaST uses a specially designed GAN to train the alternative model. The author team attacked online machine learning models on Microsoft Azure, using their method, and the remote model incorrectly classified 98.35% of the adversarial samples. Compared to algorithms that rely solely on input images to generate adversarial perturbations, inspired by previous work (Mao et al. 2019), proposed a novel general-purpose frame GAP++ (Poursaeed et al. 2018) based on GAP, which can infer the target from input images and target label conditions. Although GAP++borrows the network architecture and normalization techniques from the original GAP, its performance is lighter than GAP, making it suitable for various attack tasks. The standard evolutionary strategy (ES) algorithm can perform black-box attacks, where the Gaussian distribution is widely used as the search distribution. However, it may lack flexibility to capture the different distributions of adversarial perturbations around different benign examples.

The evolution of the above-mentioned adversarial attack algorithms reflects the design philosophy of researchers constantly seeking a balance between attack efficiency, conceal-

ment, transferability, and applicable scenarios. We pay special attention to several key trade-offs: the trade-off between the efficiency of white box attacks and the practicality of black box attacks; The cost between the speed of single step attacks and the high success rate of iterative attacks; And the contradiction between the accuracy of specific disturbances and the broad applicability of general disturbances. These internal trade-offs determine the different positions of different methods in theoretical research and practical applications.

Table 7 lists the above typical adversarial attack methods from the aspects of attack type, attack target, attack frequency, disturbance type, norm, and attack strategy.

Based on the conditional luminescence model and Gaussian distribution, Mao et al. (2020) proposes a new distribution search evolutionary strategy condition glow-based evo-

Table 7 Representative adversarial attack methods in chronological order

Attacks name	Attack Type	Target	Frequency of attacks	Disturbance	Perturbation norm	Attack strategies
L-BFGS	White box	Have goals	One step	Specific perturbations	I_∞	Constrained optimization
FGSM	White box	Have goals	One step	Specific perturbations	I_∞	Gradient optimization
JSMA	White box	Have goals	Iteration	Specific perturbations	I_2	Sensibility analysis
C&W	White box	Have goals	Iteration	Specific perturbations	$I_0 I_2 I_\infty$	Constrained optimization
One-Pixel	Black box	Driftlessness	Iteration	Specific perturbations	I_0	Decision boundary estimation
DeepFool	White box	Driftlessness	Iteration	Specific perturbations	$I_0 I_2 I_\infty$	Gradient optimization
ZOO	Black box	Have goals	Iteration	Specific perturbations	I_2	Migration calculations
UAP	White box	Driftlessness	Iteration	General disturbance	$I_2 I_\infty$	Gradient optimization
AdvGAN	White box	Have goals	Iteration	Specific perturbations	I_2	Generative model
UPSET	Black box	Have goals	Iteration	General disturbance	I_∞	Gradient approximation
ANGRI	Black box	Have goals	Iteration	Specific perturbations	I_∞	Gradient approximation
Houdini	Black box	Have goals	Iteration	Specific perturbations	$I_2 I_\infty$	Constrained optimization
BPDA	Black box	Have goals	Iteration	Specific perturbations	$I_2 I_\infty$	Gradient approximation
DaST	Black box	Have goals	Iteration	Specific perturbations	I_∞	Generative model
GAP++	White box	Have goals	One step	General disturbance	$I_0 I_2 I_\infty$	Generative model
CG-ES	Black box	Have goals	Iteration	Specific perturbations	$I_0 I_2 I_\infty$	Evolution strategy
ATNs	White box	Have goals	Iteration	Specific perturbations	I_∞	Generative model

Each method is characterized by its attack type (white-box/black-box), target, frequency, perturbation type, norm constraints, and underlying strategy

lution strategy (CG-ES) to address the problem of score black-box attacks. CG-ES transforms Gaussian distribution variables into another space using a conditional flow-based model, enhancing the ability and flexibility to capture the inherent distribution of adversarial perturbations on benign examples. Additionally, Feng et al. (2020) suggest pre-training the c-Glow (Lu et al. 2020) model by approximating the energy-based model with the perturbation distribution of the alternative model. The pre-trained c-Glow model is then initialized as the attack target model in the evolutionary strategy (ES). Thus, the proposed CG-ES method combines query-based and transfer-based attack methods, achieving higher attack success rates and efficiency. Other focuses on improving the generalization ability of models that generate adversarial samples, thereby enhancing their transferability (Wu et al. 2024). The authors combine two components: Sharp-Aware Self-Distillation (SASD) and weight scaling to improve attack transferability. The emergence of LLMs has led many research teams to focus on their robustness. Domico et al. (2025) implemented an effective black-box escape attack on large models using reinforcement learning.

However, the effectiveness of attack algorithms is not absolute and highly dependent on the infrastructure and inductive bias of the target model. This' architecture specificity 'is particularly evident in Vision Transformers. Early research proposed new hypotheses and challenges regarding the robustness of ViTs and CNNs based on their fundamental differences in feature extraction mechanisms (self attention vs. local convolution).

The advent of ViTs has prompted a dedicated line of inquiry into their adversarial robustness. Initial findings suggested that the self-attention mechanism of ViTs might confer inherent robustness advantages over traditional Convolutional Neural Networks (CNNs) (Mahmood et al. 2021). However, subsequent research has revealed that ViTs are highly susceptible to specialized attacks that exploit their unique architectural features, such as the attention mechanism and token-based representations. These include attacks utilizing adaptive token tuning (Ming et al. 2024), feature diversity (Gao et al. 2024), enhanced transferability via token gradient regularization (Zhang et al. 2023a, b, c, d) or other strategies (Guo et al. 2024a, b; Fu et al. 2022). In response to this, novel defense strategies have been explored. Beyond adversarial training variants, one promising direction involves designing more stable optimization dynamics, such as incorporating Lyapunov-stable equilibrium points into the training process (Kang et al. 2021). This growing body of work underscores that robustness is not an intrinsic property of an architecture but a complex attribute that must be designed for and assessed through specialized evaluations.

The research on attack and defense of ViTs reveals a principle: adversarial vulnerability is not an isolated attribute, but is closely coupled with the computational graph structure, training methods, and feature representations of the model. This explains why attacks designed for CNN (such as gradient based FGSM/PGD), may fail when directly migrated to ViT, leading to targeted strategies such as adaptive token tuning and feature diversity attacks. Throughout this section, the development of adversarial attacks has always revolved around the cycle of "hypothesis exploitation adaptation": attackers first make assumptions about the model (such as accessible gradients and linear decision boundaries), and design methods to exploit them; Once the defending party changes the model (such as using ViT for adversarial training), the attack hypothesis is no longer valid, driving a new round of innovative attack algorithms. We believe that there are several key limitations to current research: firstly, most attacks still focus on the digital domain and rely on simplified norm constraints, which are far from the complex perturbations in the physical world such as lighting and

viewpoint changes. Secondly, attacks on discrete spaces are still in their early stages and lack mature theoretical and tool systems like in the field of images. Finally, and most importantly, the essence of transferability is still unknown. Although it is known that integration attacks, input transformations, and other methods can improve transferability, their theoretical foundations are weak, making it difficult to guarantee the reliability of black box attacks.

4.2.2 Adversarial training

Adversarial Training (AT) is a method designed to enhance the robustness of AI models against adversarial attacks (Goodfellow et al. 2014). The goal of adversarial training is to improve the network's ability to handle adversarial samples through training, rather than relying on data preprocessing or prior knowledge. Depending on the number of iterations and whether an auxiliary model is used, adversarial training can be categorized into single-step adversarial training, multi-step adversarial training, and integrated adversarial training.

Madry et al. (2017) introduced the first influential adversarial training method, MadryAT, which addresses a Min-Max optimization problem. The introduction of MadryAT has attracted numerous researchers to explore the role of adversarial training in defense against attacks. Allen-Zhu and Li (2022) explains why adversarial training is effective, where the author argues that adversarial training purifies the learned features. Currently, adversarial training is considered one of the most effective and promising methods among all defense techniques.

In the other work, researchers examines the robustness of pre-trained vision transformers in downstream visual tasks and introduces a data augmentation virtual adversarial training method called MixVAT to enhance their resistance to adversarial attacks (Shi et al. 2022). By jointly optimizing the objective function on unlabelled data, MixVAT not only leverages additional unlabelled data to boost model robustness but also maintains a good balance with standard accuracy. Zhang et al. (2023a, b, c, d) focuses on the robustness of unsupervised domain adaptation (UDA) models, particularly their ability to handle adversarial attacks. To enhance the robustness of UDA models, the authors propose an adversarial alternating training (AAT) algorithm based on GAN. Experimental results indicate that AAT can significantly enhance the model's robustness while maintaining high clean data accuracy, compared to standard training and traditional adversarial training methods. Fakorede et al. (2023) addresses the issues with traditional adversarial training and its variants, analyzing how to integrate the hyper-sphere embedding method (Pang et al. 2020) into adversarial training to enhance the model's adversarial robustness. The author introduces a new training objective Angular-AT, which integrates HE into AT by adding two regularization terms. Experimental results show that Angular-AT significantly boosts the model's robustness on CIFAR10/100 and TinyImageNet datasets, demonstrating the effectiveness of the proposed method. Hamidi et al. (2024) proposes a novel deep neural network training method that enhances model robustness by limiting the output range caused by norm edge perturbations added to clean samples, known as adversarial polyhedra.

Most studies focus on adversarial robustness defense measures for image models, but some work focuses on the field of natural language processing (Chen et al. 2024). By integrating adversarial training with multi-layer graph neural networks, the study aims to enhance model performance, adversarial robustness, and explainability. The authors use BertSum to effectively manage and extract information from long articles and employ

various graph neural network models to process complex text data. Kuang et al. (2024) identifies issues with current adversarial training methods, which only handle individual training samples and overlook the underlying topological structure of the training data. Yang et al. (2024) proposes a method for adversarial training of diffusion models: structure-aided adversarial training (SADM).This method explained through a joint sample diffusion model, can predict samples under the original distribution, thereby enhancing the quality of generated images. Fang ‘s study examines the robustness of LLMs against various types of retrieval noise and introduces a new training method, Retrieval-Enhanced Adaptive Adversarial training (Fang et al. 2024a, b). Jia et al. (2024) research identifies that standard fast adversarial training is prone to overfitting and proposes a new fast adversarial training (FAT) method that generates initializations using high-quality attack samples and model weights from historical training processes, and improves by using buffer gradients, effectively preventing overfitting. Some papers integrate adversarial training with other methods to further enhance the model's robustness. Lin et al. (2024) enhances the robustness and generalization of the model by integrating adversarial training-optimized purifier models with random transformations. With the rapid advancement of generative LLMs, researchers have also.

explored adversarial training methods for these models. In the line, two adversarial training algorithms for LLMs: continuous adaptive training and continuous adversarial IPO are proposed (Xhonneux et al. 2024). Compared to traditional adversarial training methods based on discrete attacks, these two methods significantly reduce computational costs by calculating attacks in a continuous embedding space, making them particularly suitable for models with a large number of parameters.

Adversarial training has evolved from a heuristic technique to a subfield with rich algorithm variants in the past few years. However, our analysis reveals that it still faces several fundamental challenges: firstly, the trade-off between robustness accuracy and efficiency robustness seems to be inherent in the current AT paradigm based on empirical risk minimization. This suggests that we may need to break out of the Min Max framework and seek new definitions and optimization objectives for robustness. Secondly, the vast majority of AT methods optimize for a pre-defined and simplified threat model. This' training oriented 'characteristic leads to poor performance of the model when facing out of distribution attacks such as different norms, semantic perturbations, and physical world attacks. Future ATs need to evolve towards more universal and adaptive threat models. Despite successful practice, there is still a lack of solid theoretical explanation for why AT is effective and how its generalization performance is guaranteed. This limits our ability to design new methods and diagnose the reasons for failure.Interestingly, similar trade-offs between robustness and performance have been observed in other domains beyond adversarial training. For instance, in the context of distributed learning systems, Xue et al. (2025a, b) demonstrated a fundamental trade-off between privacy protection level (achieved through noise injection) and system convergence performance. This parallel suggests that the robustness-accuracy dilemma may be a universal challenge across different AI subfields, and insights from one domain (e.g., adversarial training) could potentially inform solutions in another (e.g., privacy-preserving distributed learning).

4.3 Robust enhancement strategy

The robust reinforcement strategies for AI models typically focus on three aspects: input data augmentation, optimization of training strategies, and model enhancement. In the previous sections, we have reviewed methods for optimizing training strategies from the perspectives of adversarial attacks and adversarial training. This section will provide a comprehensive summary of the latest research findings from the broader technical approaches of input data augmentation and model enhancement.

4.3.1 Input data enhancement

Data augmentation is an effective strategy to enhance the robustness of models. During the data collection and preprocessing stages, it artificially increases the diversity of the training set to simulate various potential variations in the test set, enabling the model to learn more generalized feature representations. Most studies focus on adversarial robustness data augmentation, using existing data or adversarial samples to expand the original training dataset. However, many researchers are also working on non-adversarial robustness data augmentation methods, such as noise (Qian et al. 2019), non-adversarial perturbations (Laugros et al. 2020; Ning et al. 2021; Zhong et al. 2022), false correlations (Chang et al. 2021; Wang and Culotta, 2021), and distribution shifts (Zhang et al. 2022a, b), which can also reduce the model's robustness and performance.

In (Mohammad et al. 2024), the author addresses the shortcomings of intrusion detection systems by employing data preprocessing and augmentation techniques. The author normalizes numerical columns, balances the dataset using oversampling techniques, and generates new samples through synthetic minority over-sampling techniques. Multi-label classification faces issues of imbalance and noise; Song et al. (2024) introduced a balanced hybrid method that uses two samplers to generate diverse minority augmented instances and manages multi-label labels meticulously. Iqbal et al. (2024) combines transfer learning with data augmentation methods, weakening the features of face image data through random rotation, scaling, and other techniques, thereby enhancing the ability to detect deep fakes. This type of traditional enhancement improves the robustness of the model to basic changes by introducing simple and usually semantically preserving transformations, which has the advantages of low computational cost and easy implementation. However, its enhancement space is limited, making it difficult to simulate complex and semantic level distribution changes, and its role in improving adversarial robustness is relatively limited. Fang et al. (2024a, b) focuses on image data augmentation methods for controllable diffusion models, using visual priors to guide the generation of synthetic images, and then guiding the diffusion model through carefully constructed prompts to ensure that the generated images are associated with specific categories. Post-filtering techniques are introduced to further improve data quality. Islam et al. (2024)also focus on data augmentation methods for diffusion models, first guiding the diffusion model to generate synthetic images related to the original images, then stitching these with parts of the original images to form hybrid images, and finally introducing post-filtering techniques to increase the diversity of training data by randomly selecting self-similar hybrid images. Some work (Wang et al. 2025a, b) trains a control network to generate synthetic images under various densities and scenarios using a pre-trained stable diffusion model. Wang et al. (2024a, b) focused on data augmentation

techniques in speech recognition, proposing speed perturbations and the use of GAN to simulate various types of speech signals, thereby increasing the diversity of training data. Wen et al. (2024) introduced methods for enhancing time series datasets, experimenting with a highly imbalanced dataset of solar flare predictions. By adding Gaussian noise to the dataset's overall statistical features, they generated synthetic data, which improved the representation of minority classes. Generative models can synthesize highly realistic and diverse data, greatly expanding the enhanced semantic space. However, such methods heavily rely on the quality and generalization ability of pre-trained generative models, and the synthesis process is computationally expensive. More importantly, it may inadvertently introduce biases or artifacts within the generative model itself, which, if not identified, could lead to the model learning false correlations. In recent years, the emergence of LLMs has enabled the natural language processing field to leverage the powerful language generation capabilities of LLMs to rewrite original training samples conceptually similar but semantically different (Dai et al. 2025), thus expanding the training dataset, even in the absence of high-quality and diverse datasets. Xue et al. (2024) proposed a data distillation method, applying geometric regularization to enhance data robustness against adversarial attacks. They not only aimed to maintain the classification accuracy of the original datasets but also enhanced the model's robustness to adversarial perturbations by minimizing the curvature of the loss function. Enhancing the semantic understanding ability of LLM can generate semantically coherent variants, especially suitable for natural language processing tasks. But the quality of its generated results is uncontrollable, which may introduce new errors or biases, and relies on black box LLM, making its enhancement effect difficult to reliably predict.

The core demand for input data augmentation is to expand the training distribution at a limited cost to cover potential distribution shifts that may be encountered during testing. However, there is a fundamental contradiction in its effectiveness: enhancement should maximize diversity to enhance generalization, while also ensuring that the enhanced samples remain true and effective in the target task space. We believe that the current key issues include: ensuring semantic consistency. For complex modalities such as images and language, simple geometric or lexical substitutions may disrupt semantics, while complex generation methods are difficult to ensure controllability; Enhanced data, especially synthetic data, may deviate from the true data distribution, leading to overfitting of the model on enhanced data and decreased performance on real data; Adversarial enhancement has limitations, as data generated through adversarial attacks can selectively expose model weaknesses, but it tends to simulate worst-case scenarios and may not cover more common natural disturbance patterns. From the perspective of research trends, the future direction may lie in the development of theory guided reinforcement strategy selection theory, which automatically selects the most effective reinforcement combination based on model capacity, task characteristics, and available data scale. And research on provable enhancement techniques to ensure that the enhancement does not change the semantic labels of the samples, construct an enhancement evaluation loop, and dynamically adjust the enhancement strategy based on the performance of the model on the enhanced data.

4.3.2 Model enhancement

Another strategy to improve the robustness of AI models is to modify the overall architecture or part of the network layer structure of the model, so that it can more easily capture the

focus features of input data, thus improving the robustness of the model. Most paper divides the method of model enhancement into two parts: model architecture optimization and post-processing strategies (Table 8).

4.3.2.1 Model architecture optimization There are numerous methods for optimizing model architectures, and different modification strategies are applied to various original network structures. The most common approach is to incorporate attention modules into the original network. Appropriate spatial and channel attention modules can help the model capture the required input features (Zhang et al. 2025). Additionally, many studies focus on developing more robust loss functions and regularization techniques, which can enhance the model's robustness to varying degrees (Li et al. 2025). The attention module can enhance the robustness to key information by dynamically focusing the model on important features. But the attention map itself may become a new target of attack, and adversarial perturbations may be designed to mislead the attention distribution.

Liang et al. (2022) introduces a method to enhance model decision-making by suppressing the logit values of the highest confidence classes in neural networks, thereby improving the model's robustness against adversarial gradient attacks. This approach integrates with existing adversarial training frameworks like TRADES, enhancing the model's defense against adversarial attacks without significantly increasing computational costs. Liu et al. (2024a, b) proposes a defense strategy for no-reference image quality assessment (NR-IQA) models, aimed at improving the model's robustness against adversarial attacks. The authors demonstrate through theoretical analysis that changes in the model's prediction scores are closely related to the L_1 norm of the gradient of the input image. Based on this, they propose a training strategy to enhance the model's robustness by regularizing the L_1 norm of the gradient. Several studies focus on developing new loss functions to enhance the defense against adversarial attacks in a AI models. In the line, Zhang et al. (2024) introduces a loss function called Weight Decorrelation to achieve defense enhancement, aiming to disrupt the ordered structure in the feature space, thereby improving the network's robustness. The

Table 8 Recent data enhancement methods for improving model robustness

Thesis	Modality	Experimental prototype	Data set
Mohammad et al. 2024	Image	CNN	UNSW-NB15, etc.
Song et al. 2024	Image	CNN	ImageNet, etc.
Fang et al. 2024a, b	Image	Diffusion model	PASCAL VOC
Wang et al. 2024a, b	Phonetic	Wav2vec2.0	UASpeech, etc.
Islam et al. 2024	Image	Diffusion model	Flower102 et al.
Iqbal et al. 2024	Image	CNN	190,335 real and fake images
Wang et al. 2025a, b	Image	diffusion model	ShanghaiTech, etc.
Dai et al. 2025	Text	LLM	Amazon Dataset
Wen et al. 2024	Time series	/	SWAN-SF
Xue et al. 2025a, b	Image	ResNet18	ImageNet, etc.

The table summarizes techniques by modality, experimental prototype, and datasets used for validation

authors draw on principles from physical chemistry, where increasing the disorder in a system enhances its energy stability, making it harder to disrupt the ordered state of the system. These loss functions encourage robustness by explicitly constraining feature representations or optimization processes, providing a fundamental improvement approach. However, its design often relies on specific assumptions about the sources of vulnerability (such as feature correlation leading to vulnerability), and may introduce additional hyperparameter tuning burdens. Guo et al. (2022) found that sub-networks with weak robustness are significantly correlated with the overall low robustness of the network when analyzing the robustness issues within the neural networks of AI models. They proposed a new search algorithm to identify weak sub-networks and enhanced their performance through knowledge distillation. This method can be integrated with existing data augmentation and adversarial training techniques to further improve model robustness. Wang et al. (2021) examined how network architecture affects robustness in image style transfer, finding that residual connections lead to the dominance of low-level features, which significantly degrades the performance of style transfer. By using Softmax activation smoothing, they softened the feature activations in the loss function, thereby enhancing the robustness of various network architectures, including those with random initialization and pre-trained models, in style transfer tasks. In the other work, a stable heat map regression method to address the robustness challenges in image recognition was introduced (Zhang et al. 2022a, b). This method improves key point recognition and simplifies optimization by incorporating three core modules: row-column correlation layers, heat map regression, and maximum stability training loss functions, ensuring the model's stability even when dealing with distorted images. Neural architecture search (NAS) is a method for automatically discovering the optimal neural network architecture, and many studies have applied this approach to enhance model robustness. Devaguptapu et al. (2021) investigated the impact of parameter quantity changes on model adversarial robustness and found that NAS can replace costly adversarial training. Beyond neural architecture modifications, the development of novel optimization algorithms also contributes to model robustness in complex tasks. For example, Huang et al. (2024a, b, c) proposed a Nonlinear Activated Beetle Antennae Search Improving Egret Swarm Optimization Algorithm (NBESOA) to solve the constrained mean-variance portfolio selection problem. The algorithm's ability to handle non-convex, multi-constraint optimization demonstrates its potential for enhancing robustness in financial decision-making systems. Mok et al. (2021) focuses on the use of NAS methods to strengthen NAS methods by introducing different regularization forms in aspects such as network sensitivity; Li et al. (2021) integrates NAS into existing models to find the minimum model parameter increase capacity that allows the model to withstand adversarial attacks. NAS automates the discovery process of robust architecture, demonstrating enormous potential. However, the design of its search space highly relies on prior knowledge, and the huge computational cost makes it difficult to widely reproduce and apply. The search results may also be overfitting to specific datasets and threat models.

4.3.2.2 Post-processing strategy Using post-processing strategies can also significantly enhance the robustness of AI models. One of the most common methods is input anomaly detection, which involves adding a simple neural network to the input layer of the original model to filter out obviously unreasonable input data. Input purification, on the other hand, involves analyzing the gradient information of the input data to identify and correct

perturbations, thereby purifying the data to obtain clean samples. Zhang et al. (2023a, b, c, d) summarized the relevant methods for detecting and purifying input data, leveraging the complementary nature of purifiers and detectors to propose a cooperative mechanism. The purifier first processes the input data and removes interference items; if the input has not been processed, the detector will capture any abnormal behavior in the sample. Additionally, the detector can add noise to confuse the direction of attackers, making it difficult for them to generate effective attack samples, thus greatly enhancing the robustness of AI models. However, this type of method has the advantages of non invasiveness and plug and play, making it suitable for deployment on existing systems. But the purifier or detector itself constitutes a new and potentially more vulnerable module, and attackers can launch targeted attacks.

Defense distillation is a method that uses knowledge distillation to enhance the model's defense against adversarial attacks (Papernot et al. 2016a, b). By adjusting the temperature of the softmax layer output from the teacher network, soft labels are generated to train the student model, enabling it to learn more information and thus enhancing robustness. Similarly, Sarkar et al. (2024) demonstrated that defense distillation effectively protects against various adversarial attacks, notably improving the model's robustness against the FGSM attack. However, its effectiveness in mitigating C&W attacks was significantly reduced.

Model pruning algorithms are designed to reduce the cost of training models by compressing their size (Li et al. 2024). Chen et al. (2022) explored a heuristic pruning algorithm to enhance the robustness of AI models. They selectively replaced ReLU neurons deemed useless with linear activation functions, which helped improve robustness at minimal performance cost. Ensemble learning improves overall model performance by combining the results of multiple models, thereby reducing the risk of overfitting in a single model (Mienye et al. 2022). Waghela et al. (2024) introduced a dynamic ensemble learning method that leverages model diversity to dynamically adjust the ensemble configuration based on input features and detected attack patterns, thus enhancing the overall robustness of the model. The integrated approach inference cost is directly proportional to the number of models, which may not be feasible in real-time systems, and how to ensure sufficient diversity of the base model itself is a challenge.

Model enhancement attempts to internalize robustness into the computational process of the model or ensure it through external mechanisms. However, both paths face internal limitations. The fundamental challenge for architecture optimization lies in the contradiction between robustness and expressive power. Increasing the capacity and nonlinearity of the model (such as deeper networks and attention mechanisms) can usually enhance its expressive power, but it may also introduce more complex decision boundaries and more fragile points at the same time. Most architecture modifications lack interpretable robustness attribution, making it difficult to determine whether the performance improvement is due to the structure itself or an accidental combination of hyperparameters. The core dilemma of post-processing strategies lies in the additivity of defense and system complexity. Placing defense outside the model, although flexible, also exposes defense modules as independent attack surfaces. At the same time, this leads to system level complexity: unexpected interactions may occur between the model, purifier, and detector, making it difficult to analyze and verify the overall system behavior.

5 Robustness of large language models

LLMs represent a prominent and rapidly evolving subclass of deep learning-based AI models. Characterized by the Transformer architecture, massive parameter scales (often billions to trillions), and pre-training on vast textual corpora, they have demonstrated exceptional capabilities in generative and understanding tasks. While many fundamental principles of robustness discussed in previous sections (e.g., adversarial attacks, data augmentation) apply to LLMs, their unique architecture, generative nature, and failure modes—such as the hallucination phenomenon—necessitate specialized analysis. In this section, we focuses on the robustness challenges and enhancement techniques specific to LLMs, situating them within the broader AI robustness landscape established earlier. We describe the Major robustness threats in LLMs in Table 9.

As the parameters of AI models continue to grow, generative LLMs like GPT have emerged. The massive number of parameters makes the training process extremely complex (Achiam et al. 2023). To avoid hallucinations caused by overly large training datasets (Huang et al. 2025), researchers are constantly exploring new methods. From early prompt templates (White et al. 2023) to later joint intelligences (Zhao et al. 2024a, b), efforts have been made to enhance the robustness of LLMs in various ways. Although there are differences between LLMs and traditional models, the potential specific robustness threats can be categorized into three main areas: hallucinations, lack of context understanding, and the absence of reasoning and causal relationships (Howe et al. 2024). Most existing paper focuses on these aspects, exploring how to enhance robustness from one or more of these angles.

In this section, we first review the causes and mitigation methods of hallucination phenomenon in LLM, and then discuss how to evaluate and strengthen the robustness of LLM from other different perspectives.

5.1 LLM robustness enhancement: hallucination relief

The hallucination problem in LLMs refers to the phenomenon where models may produce information that is inconsistent with reality, illogical, or entirely fictional when generating text. This typically occurs when the model attempts to answer uncertain questions or make inferences without sufficient context. According to (Carlini et al. 2021), the hallucinations

Table 9 Major robustness threats in LLMs

Robustness threat	The cause of the problem
Hallucinations: factual hallucinations and faithful hallucinations	LLM generates text based on probability, and does not guarantee the authenticity of the generated content.
Lack of contextual understanding	The complexity of natural semantics and the problem of long distance dependence
Lack of reasoning ability and causality	LLM is based on pattern matching rather than logical reasoning
Data bias and noise	The training data set has bias or noise, and the model cannot learn accurate knowledge
Model complexity and overfitting	The large amount of training data of LLM leads to serious overfitting phenomenon

The table identifies four primary threat categories (hallucinations, lack of contextual understanding, reasoning limitations, data bias) and analyzes their root causes

Table 10 Comprehensive analysis of LLM hallucination causes and mitigation strategies

The cause of hallucinations		Methods to relieve hallucinations	
Data hallucination	Data errors and bias	Data augmentation	Data filtering
	Knowledge boundaries		Model editing
	The data is out of alignment		Search to enhance generation
Train hallucination	Pre-training hallucinations	Training optimization	Reduce pre-training illusions
	Fine-tuning hallucinations		Reduce the illusion of alignment
	RLHF megrim		Post-training methods
Infer hallucination	Decoding strategy issues	Decoding optimization	Enhance factual decoding
	The reasoning failed		Enhance fidelity decoding

Hallucinations are categorized by origin (data, training, inference) with corresponding relief methods for each category

in LLMs differ from those in traditional natural language processing and can be primarily categorized into two types: factual hallucinations and fidelity hallucinations. Factual hallucinations highlight discrepancies between the generated content and verifiable real-world facts, often manifesting as inconsistencies. Fidelity hallucinations capture deviations between the generated content and user input or internal contradictions within the generated content. Similar to non-LLM AI models, the causes and mitigation methods of LLM hallucinations can be classified into three aspects: data augmentation, training optimization, and inference decoding optimization. Table 10 outlines the causes and mitigation methods of LLM hallucinations.

5.1.1 Data hallucinations

The data used for training LLMs consists of two primary components: pre-training data, which enables LLMs to acquire general capabilities and factual knowledge; and aligned data, which teaches LLMs to follow user instructions and align with human preferences. This is primarily reflected in three aspects: the presence of false information and biases in incorrect pre-training data sources, the inherent knowledge boundaries within the scope of pre-training data, and the illusions caused by unaligned data.

AI models have an inherent tendency to remember training data, and this memory tendency increases as the model grows in size. Generally, in the process of generating hallucinations, the inherent memory capability acts as a double-edged sword. While LLMs can capture more information due to their memory capabilities, they are also susceptible to being misled by false or biased information in pre-trained data. Misleading information, such as fake news and rumors, is one of the primary reasons for LLM hallucinations. Due to their powerful memory, LLMs are more likely to produce incorrect outputs compared to traditional AI models, leading to imitative falsehoods. Although large-scale pre-training datasets input vast amounts of knowledge into LLMs, there are still knowledge boundaries.

This is because LLMs cannot remember all the information they encounter, especially long-tail knowledge. Additionally, the pre-training data itself has its own knowledge boundaries. When LLMs encounter queries that exceed their knowledge boundaries, hallucinations are more likely to occur. Misaligned data is another reason for LLM hallucinations. In this work, Gekhman et al. (2024) analyzed the training dynamics of integrating new factual knowledge during supervised fine-tuning and found that LLMs struggle to effectively acquire such new knowledge. They discovered a correlation between acquiring new knowledge and increasing hallucinations, indicating that introducing new factual knowledge enco-urages LLMs to generate hallucinations. Furthermore, Li et al. (2024a, b) found that overly complex and diverse instructions can also lead to more hallucinations.

The methods to alleviate hallucinations caused by data in LLMs are primarily categorized into three types: data filtering, model editing, and retrieval-enhanced generation. Data filtering involves carefully selecting high-quality data for the training dataset and manually excluding low-quality, misaligned, or biased data. However, as pre-trained datasets grow in size, the academic community is increasingly focused on how to collect high-quality data. Data filtering d--irectly addresses the root cause of hallucinations, making it the most effective and straightforward strategy (Li et al. 2023a, b, c). Model editing involves introducing additional knowledge to correct the model's erroneous behavior, with two main approaches: post-editing and meta-learning (Meng et al. 2022). Post-editing identifies the parameters in the LLM that are causing errors and updates them using specific methods to alter the LLM's output behavior. Meta-learning trains an additional neural network to predict the weight updates of the LLM, which incurs a higher cost. Retrieval-augmented generation (RAG) (Shuster et al. 2021) first retrieves relevant knowledge from external sources and decouples it from the LLM. RAG can effectively mitigate hallucinations caused by knowledge gaps without impacting the LLM's performance. Commonly used methods in RAG include one retrieval, iterative retrieval, and post-processing retrieval. The advantage of RAG is its versatility, applicable across various fields, but it is susceptible to irrelevant retrieval.

The intervention at the data level is based on a core assumption: hallucinations mainly stem from errors or missing in the training data. However, there are two fundamental issues with this assumption. Firstly, what is' incorrect 'data? The boundary between facts and opinions, universal consensus and emerging knowledge is often vague and dynamic. Overly strict filtering may exclude valuable controversial content or cutting-edge explorations, leading to the rigidity of the model's knowledge system. Secondly, knowledge is dynamic. Model editing and RAG attempt to introduce external knowledge to compensate for the static nature of training data, but the scale of model editing is limited, while the effectiveness of RAG relies entirely on the accuracy and timeliness of the retrieval system, and introduces additional system complexity and latency. It is worth noting that even if the data is 'completely correct ', the model may still produce illusions. This is because memory does not equal understanding, and association does not equal causality. The model may perfectly remember the facts in the training data, but cannot correctly combine and apply them in inference. Therefore, simple data correction is difficult to touch upon the cognitive roots of hallucinations.

5.1.2 Training hallucinations

The training hallucinations of LLMs encompass pre-training hallucinations, fine-tuning hallucinations, and reinforcement learning from human feedback (RLHF) hallucinations based on human knowledge. Pre-training leverages large datasets to equip LLMs with fundamental knowledge. Typically, LLMs are structured around transformers, and during the pre-training phase, they use causal language modeling, which operates in a unidirectional left-to-right manner, allowing the model to learn and predict subsequent outcomes. This approach inherently limits the model's ability to capture complex contextual dependencies, potentially increasing the risk of hallucination generation (Li et al. 2023a, b, c). Researchers have also observed that LLMs can sometimes capture long-range and short-range dependencies, leading to unpredictable hallucinations (Gekhman et al. 2024). As the sequence length increases, the attention mechanisms in the model architecture become sparser. Supervised fine-tuning is used to adapt LLMs for downstream tasks, but this often leads to severe overfitting by training LLMs to learn content beyond their knowledge boundaries, thereby exacerbating hallucination generation. RLHF hallucinations, based on human feedback, arise when LLMs are over-trained, leading them to cater to user preferences at the expense of authenticity (Cotra et al. 2021).The causes of training hallucinations typically stem from the limitations in the architecture and training strategies of LLMs. Researchers have explored methods to address these issues by examining the three aspects that lead to training hallucinations. To reduce pre-training hallucinations, researchers aim to improve the unidirectional prediction and attention sparsity in the original model training. Liu et al. (2024) proposed the potential of encoder-decoder models to leverage their context window capabilities. Bidirectional autoregressive methods enable the model to consider both past inputs and future outputs, thus capturing dependencies from two perspectives.

One method to address the illusion of supervision fine-tuning is to define the precise knowledge boundaries of LLMs, which can bridge the gap between the knowledge presented in LLM memory and labeled data. Sharma et al. (2023) found that aggregating multiple human preferences can enhance the quality of LLM outputs, thereby reducing the illusion caused by RLHF. Moreover, post-training methods, which bypass the complexities of data sources and pre-training, can also alleviate training illusions.

The enhancement during the training phase attempts to shape the intrinsic cognitive structure of the model by changing the learning objectives. This touches upon the core contradiction of the current LLM paradigm: the model is trained to predict the next word in a sequence (a statistical task), but we expect it to perform logical and factual reasoning (a cognitive task). There is an inherent deviation between the causal modeling objectives in the pre training phase and the factual generation objectives. Introducing bidirectional encoding or improving attention mechanisms may alleviate long-range dependency problems, but cannot fundamentally endow models with the ability to verify facts or perform deductive reasoning. Supervised fine-tuning (SFT) and human feedback based RLHF can inject human preferences into the model, but they are prone to introducing new biases: alignment taxes. Models may learn to generate "safe but hollow" text in order to obtain high rewards, or mechanically mimic the style of human annotators rather than deeply understanding the content. A deeper dilemma is that we expect the model to have both powerful memory and reproduction capabilities, as well as creative reasoning abilities beyond training data. These

two abilities have inherent tension in the current training paradigm, and over optimization of one often comes at the expense of the other.

5.1.3 Inferential hallucinations

The performance and robustness of LLMs in the inference stage are primarily determined by their decoding methods. Most LLMs currently use a decoding strategy based on probability sampling (Holtzman et al. 2019). The reason for introducing randomness into this strategy is that sequences with high probabilities often result in low-quality text, a phenomenon known as the likelihood trap. The diversity of decoding strategies gives LLMs their remarkable generative capabilities, but it also increases the risk of hallucinations (Zhang et al. 2021). The process of LLMs generating predictions depends not only on context but also on parts of the generated text. Additionally, due to the locality of the attention mechanism, LLMs have a higher risk of forgetting instructions, leading to fidelity hallucinations where the output deviates from the context (Shi et al. 2024). In contexts where multiple questions are asked simultaneously, even if LLMs possess necessary knowledge, they may fail to recognize the connections between questions due to a lack of sufficient reasoning ability (Zheng et al. 2023), resulting in hallucination phenomena. Enhancing the authenticity and fidelity of decoding is an effective method to reduce the illusion of reasoning generated by LLMs.Authenticity-enhanced decoding aims to improve the reliability of LLM outputs by prioritizing the authenticity of the generated information. The focus is on aligning the model's output with established real-world facts, thereby minimizing the risk of spreading false or misleading information. Lee et al. (2022) introduced a kernel sampling algorithm that dynamically adjusts the kernel probability during sentence generation, considering that the randomness in the sampling process might introduce non-factual content. By dynamically adjusting the kernel probability based on decay factors and lower bounds, and resetting the kernel probability at the start of each new sentence, this decoding strategy balances the generation of factual content with maintaining output diversity. Chuang et al. (2023) explored how to enhance the factuality of LLM decoding from the perspective of storing factual knowledge. Dhuliawala et al. (2024) leveraged the self-correction capability of LLMs to improve the original generated content without relying on external knowledge bases, ultimately producing an improved revised output.

Lifelike decoding enhancement considers both contextual consistency and logical consistency. The illusion of lifelikeness often arises from LLMs' insufficient attention to the given context. Shi et al. (2024) introduced context-aware decoding, which amplifies the differences in output probabilities between contexts, encouraging LLMs to focus more on contextual information rather than over-relying on prior knowledge. Chang et al. (2023) introduced a dynamic decoding algorithm that enhances lifelikeness while maintaining diversity. This algorithm involves two parallel decoding steps: one with context and one without. During decoding, the separation between the two output distributions serves as a guiding signal, indicating the relevance of the source context. Lei et al. (2023) detected contextual illusions at the sentence and entity levels and incorporated these judgments into the improvement of generated responses. Logical consistency is addressed through the concept of thought chains. To improve the logical consistency of thought chains during multi-step reasoning and logical reasoning, Wang et al. (2024a, b) adopted a knowledge distillation framework. They first used contrastive decoding to generate consistent content and then

fine-tuned the student model through counterfactual reasoning, effectively eliminating the shortcut of drawing conclusions without considering reasons. Some studies have deeply analyzed the relationship between context, thought chains, and output results during illogical reasoning, finding that the inconsistency between the results obtained by thought chains and the context is the cause (Wang et al. 2023).

The enhancement of inference strategies provides a lightweight and immediate intervention method, but its effectiveness has clear boundaries. Firstly, the decoder can only guide based on the existing probability distribution of the model and cannot correct fundamental errors in the underlying knowledge representation. If the model firmly believes in the wrong answer, no matter how clever the decoding strategy is, it will be difficult to bring it back on track. Secondly, many decoding optimization methods use specific automatic indicators as guiding signals. This may lead to overfitting of the model to these metrics, learning to generate text that scores high on the metrics but still has semantic flaws, and getting stuck in a cycle of 'metric games'. Finally, decoding hyperparameters optimized for specific tasks or datasets often fail in other scenarios. The lack of an adaptive and theoretically interpretable decoding framework makes these methods face high tuning costs in practical deployment.

Looking at the above three paths, the core dilemma of LLM illusion relief lies in our attempt to use statistical based models to accomplish tasks that require factual and logical consistency. This fundamental contradiction gives rise to three major challenges: firstly, we demand that the content generated by the model be faithful to reality, while expecting it to engage in creative writing, hypothetical reasoning, or solve open-ended problems. Both of these goals have theoretical and computational issues, and current research lacks a clear definition of "acceptable creative illusions" and "harmful factual illusions". Secondly, there are serious flaws in the evaluation paradigm. Automation indicators mainly measure surface similarity, which is disconnected from factual accuracy. Although emerging LLMs based evaluators have more potential, the evaluators themselves are also black box models, and their stability, consistency, and potential biases in judgment are elusive. Although manual evaluation is the gold standard, it is costly, difficult to scale up, and there are significant subjective differences among different evaluators. More seriously, the lack of open, comprehensive, multidimensional, and multidisciplinary standard evaluation benchmarks makes it almost impossible to make fair comparisons between different studies. Finally, the vast majority of mitigation methods are empirical improvements that lack solid theoretical guidance. We are not yet clear on what characteristics of the model representation space hallucinations manifest in, nor are we clear on which internal mechanisms of the model are altered by effective interventions. In the future, we should establish open, interpretable, and multidimensional factual evaluation benchmarks and tools, and start from the study of model interpretability to deeply explore the mechanism of hallucinations occurring within the model, in order to design more fundamental intervention measures.

5.2 LLM robustness enhancement: other perspectives

In addition to starting from the hallucination problem, many papers also discuss how to optimize prompt instruction set, multi-agent, retrieval enhanced generation and other methods to enhance the robustness of LLM. This section will explore several key directions and focus on the following questions: How do different methods respond to out of distribution

inputs? How do they balance performance improvement with computational/engineering complexity? And how effective is its defense mechanism when facing targeted attacks?

In some work, the author explores the use of chaos engineering to enhance the robustness of LLM multi-agent systems in real-world production environments (Owotogbe et al. 2025). It highlights that while multi-agents can be applied to a wide range of tasks, such as answering questions, generating content, automating customer support, and improving decision-making, they are susceptible to sudden errors or disruptions, leading to issues like hallucinations, agent failures, and communication failures. The authors propose a chaos engineering framework to proactively identify various errors in LLM multi-agents. LLMs are particularly vulnerable to instruction perturbations, and in other work, Agrawal et al. (2025) focuses on task-specific instruction editing at the character and word levels, comparing several methods: iterative self-denormalization, supervised fine-tuning self-denormalization, perplexity smoothing, instruction inheritance, and representation alignment. Iterative self-denormalization and perplexity smoothing use iterative optimization and perplexity as evaluation metrics, respectively; instruction inheritance increases model diversity by integrating multiple variants of instructions; representation alignment aims to align the hidden representations of perturbed instructions with those of unperturbed instructions. Improving the robustness of the model to instruction perturbations is crucial for practical deployment. However, current methods mostly rely on surface character or vocabulary perturbations for defense. This may result in the model remaining vulnerable to higher-level semantic deception or logical misleading attacks. In addition, whether excessive reinforcement of instruction robustness will make the model too "rigid" and lose flexible understanding of reasonable instruction variants is a trade-off issue. The authors suggest that these methods can enhance the robustness of LLMs. In the context of domain-specific applications, the design of prompts plays a critical role in the robustness of LLM outputs. For instance, in financial portfolio selection, Huang et al. (2024a, b, c) proposed a structured prompt framework comprising Strategy, Goal, and Constraints, demonstrating how the specificity of instructions influences the quality and consistency of stock recommendations generated by ChatGPT. This highlights the importance of prompt engineering as a lightweight method for improving LLM robustness in specialized tasks. Zhao et al. (2024a, b) address the robustness of LLM-generated code, noting that current code evaluation focuses on correctness but overlooks critical issues such as missing input validation and inadequate error handling. The author analyzed four of the LLMs and found that most of the generated code lacked condition checks. Based on this, the author introduced the RobGen framework, which employs two model-agnostic techniques: RobGen-Adj dynamically adjusts the probability of tags during decoding to encourage the inclusion of control structures, and RobGen-Ins inserts missing conditional statements after generation to enhance the quality of the generated code. The RobGen framework focuses on the functional security of code, such as input validation, which is a often overlooked but crucial dimension. Its method (dynamically adjusting label probabilities, inserting conditional statements afterwards) is essentially a heuristic constraint on the generation process. Its limitations lie in the fact that these constraint rules require manual pre-definition, making it difficult to cover all possible code security vulnerabilities, and may interfere with the quality of non-security related code generation. Guo et al. (2024a, b) proposed a two-stage training framework, including instruction-enhanced supervised fine-tuning and consistency alignment training. The first stage helps the model by guiding it through similar instructions, while the second

stage improves diversity by distinguishing subtle differences in similar responses and helps the model understand which responses better align with human expectations. LLMs often exhibit severe hallucination phenomena (Tao et al. 2024) when facing specific adversarial attacks, such as character-level text attacks and prison break attacks. Tao et al. conducted a comparative experiment to evaluate the robustness of LLMs against specific attacks. They randomly deleted characters from the input text to simulate common spelling errors made by users. In contrast experiments with normal prompts, they found that models such as GPT-4 and GPT-4-turbo showed a significant decrease in accuracy when dealing with such spelling errors.

In another way, the author (Zhou et al. 2025) introduces a framework called TrustRAG, designed to enhance the robustness and credibility of knowledge retrieval-based generation (RAG) systems against malicious injection attacks. TrustRAG employs a two-stage defense mechanism to filter out harmful and irrelevant content, ensuring the accuracy and reliability of the final response. Initially, the K-means clustering algorithm is used to identify potential attack patterns by analyzing the embedding distribution of documents, thereby filtering out suspicious documents. In the conflict resolution phase, the model combines its internal knowledge with external information to identify and eliminate inconsistent or irrelevant data. Finally, through self-assessment, it determines the most reliable response. Along this line, Cui et al. (2024) systematically evaluates the sensitivity of multimodal large language model (MLMM) to visual adversarial samples. Although MLMM shows lower robustness to visual adversarial inputs without additional text information, providing context related to the query can significantly improve its robustness. Specifically, in scientific question answering tasks, MLMM only experiences an 8.10% performance drop, compared to 99.73% without context. Additionally, a new image classification method for practical applications: query decomposition is proposed. This method breaks down each question into multiple inquiries about the existence of the target object and integrates relevant contextual information, further enhancing the model's robustness. In conventional randomization smoothing, the model's prediction stability is enhanced by adding noise to the input. However, this method relies on the model's performance with noisy data, and LLM often performs poorly in such scenarios. Ji et al. (2024) proposed an improved randomization smoothing approach that leverages LLM's inherent capabilities for preprocessing. This involves decoding inputs with random occlusions before performing task-related predictions. Compared to previous methods, this approach does not require additional training of a decoder and significantly enhances performance and security against adversarial attacks. These works represent efforts to enhance robustness from the perspectives of system architecture and inference processes. TrustRAG ensures the reliability of knowledge sources through multiple layers of filtering and conflict resolution, but its defense performance heavily relies on the robustness of each layer component, forming a chain dependency. Improved random smoothing methods attempt to provide provable robustness for LLM, but often at the cost of sacrificing prediction accuracy or increasing computational delay, and their theoretical guarantees still face challenges in applications in discrete text spaces.

The diverse research directions reviewed in this section indicate that the robustness of LLM has evolved from a single "output content factual" problem to a comprehensive challenge that spans the entire stack of data, algorithms, systems, and human-computer interaction. The current core issue is to enhance robustness at the component level (such as instruction robustness of a single model), system level (such as multi-agent coordination,

RAG pipeline), and interaction level (such as human-machine dialogue). However, how to ensure the robustness of higher levels with the robustness of lower levels is an under researched "combinatorial robustness" problem. A model with strong instruction robustness may still fail in multi-agent systems due to communication protocol vulnerabilities; The vast majority of methods are developed and evaluated on static datasets. However, real-world threats are dynamically evolving (such as new jailbreak prompts, constantly changing adversarial samples). The future robustness enhancement technology must have the ability of online learning or rapid adaptation to cope with the constantly emerging new attack patterns. In addition, for multimodal large models, their robustness challenges come from the fragility of visual, linguistic, and alignment processes between the two. Current research is mostly focused on single point breakthroughs, lacking a unified theoretical framework to explain and understand the impact and defense mechanisms of cross modal disturbances.

6 Discussion

This section synthesizes the insights from the preceding review to outline the persistent challenges and promising pathways forward in the pursuit of robust AI. First, we consolidate the t challenges that currently constrain the development and deployment of robust models (Sect. 6.1). Subsequently, we propose key future research directions aimed at overcoming these obstacles (Sect. 6.2). A principal conclusion drawn from this comprehensive survey is that while substantial progress has been made—particularly in adversarial robustness—the goal of achieving holistic and trustworthy robustness, which integrates seamlessly with requirements for fairness, explainability, and real-world reliability, remains an open and critical research frontier.

6.1 Challenges

6.1.1 Data diversity and distribution bias

From the majority of the paper we have gathered, it is evident that most robustness evaluation and enhancement datasets suffer from repetition issues, with a lack of large datasets and real-world scenario data. The current training data fails to cover all potential real-world scenarios, leading to a decline in the robustness of AI models when faced with unknown environments. For instance, in the field of speech recognition, Achiam et al. (2023) points out that traditional datasets often have limited templates and vocabulary spaces, causing models to over-rely on the statistical patterns in the training data, resulting in poor performance when encountering new problems. While data augmentation can increase sample diversity, the simulated data generated may deviate from the real distribution, affecting the model's generalization ability. Regarding the issue of input data noise, although adversarial attacks are widely used to enhance the model's resistance to interference, the design of perturbations to generate adversarial samples still needs to balance concealment and attack effectiveness.

6.1.2 Lack of attention to natural robustness

The excessive focus on adversarial robustness is a significant issue in the field of robustness. Most existing research focuses on how to eliminate and defend against adversarial attacks, with only a few studies addressing real-world natural robustness challenges. In visual tasks, models may perform well on carefully designed adversarial datasets but can significantly degrade under extreme natural conditions, such as heavy snow or low light. In natural language processing, spelling errors in input text can cause the model to produce severe hallucinations. These issues highlight the current AI models' lack of defense against natural robustness. Compared to structured adversarial synthetic data, simulating natural perturbations is more challenging, and the academic community still lacks a unified benchmark for evaluating natural robustness.

6.1.3 Model evaluation and architectural limitations

Most paper lacks a clear and specific definition of robustness, and the focus on robustness in the AI field varies across different scenarios. Researchers primarily focus on adversarial robustness to artificially generated adversarial samples. Currently, there is a lack of general quantification metrics for various models, making it difficult to directly compare robustness evaluation results. Existing evaluations often rely on static datasets, whereas the dynamic nature of real-world scenarios necessitates more complex dynamic evaluation frameworks.

The rapid advancement of AI is largely attributed to deep learning. However, the black-box nature of deep learning results in models that are difficult to interpret, highly sensitive to input perturbations, and prone to issues such as gradient disappearance and explosion. While techniques like regularization and normalization can mitigate these issues, they do not fundamentally address the structural vulnerabilities. The robustness challenges of LLMs and multimodal models are even more complex, as differences in information across various modalities can lead to prediction biases.

6.1.4 Dynamic game of confrontation and defense

The ongoing evolution of attack technologies necessitates that existing defense measures be continuously updated to address new types of attacks. Attackers continually exploit the vulnerabilities in current defense strategies, devising targeted attack methods. The significant disparity in resource consumption between attackers and defenders makes it challenging to maintain the security and robustness of models over the long term. Currently, most robustness enhancement methods lack rigorous theoretical support and rely more on empirical improvements. The relationship between robustness and other trustworthy AI concepts remains unclear, and the specific mechanisms linking them require ongoing exploration.

6.1.5 Robustness challenges of complex AI systems

As AI models are deployed on a large scale, increasingly complex AI systems face unprecedented robustness challenges. These models must not only ensure their own defensive capabilities but also interact with other models to maintain the system's stability and robustness. Moreover, the construction of these systems involves the coordination of multiple

software and hardware modules, and any uncertainty in one part can pose a security threat to the entire system. Therefore, when building and operating systems that incorporate complex AI models, it is essential to focus not only on the robustness of each stage of the AI model but also to review the system-level robustness from a broader perspective.

Furthermore, robustness challenges extend beyond the paradigms of adversarial and data-driven robustness discussed herein. Traditional and concurrent research in control-theoretic methods (e.g., neural ordinary differential equations for stable dynamics) and multi-agent systems addresses fundamental stability and coordination under uncertainty. While these approaches often originate from different problem settings (e.g., continuous-time control, distributed robotics), their insights into ensuring system-level reliability and resilience in dynamic environments present a valuable cross-disciplinary frontier for future trustworthy AI research.

6.1.6 Practical deployment challenges

Beyond algorithmic innovation, deploying robust AI models in real-world systems introduces paramount practical challenges that are often underexplored in academic research.

The pursuit of robustness frequently incurs significant overhead. Adversarial training, particularly multi-step methods like PGD, can multiply training time and energy consumption. Inference-time defenses, such as randomized smoothing or input purification modules, increase latency and resource demands. This creates a direct trade-off between robustness and operational efficiency, posing a critical constraint for deployment in resource-constrained environments (e.g., mobile devices, edge computing) or large-scale commercial services where throughput and cost-per-inference are key metrics.

Techniques validated on benchmark datasets may not scale effectively to real-world scenarios involving billion-parameter models (LLMs), continuous data streams, or complex, distributed systems. Furthermore, a robust model is merely one component within a larger software stack. Its resilience can be nullified if upstream data pipelines or downstream decision logic contain vulnerabilities, highlighting the need for a holistic, system-wide security and robustness assessment. Ensuring robustness across this heterogeneous compute continuum—from cloud training to edge deployment—remains an open engineering challenge.

Current academic benchmarks primarily measure accuracy under controlled perturbations but often fail to account for real-world constraints such as hardware heterogeneity, memory limits, power budgets, and real-time deadlines. There is a pressing need for industry-relevant benchmarks that evaluate robustness under these practical conditions to guide the development of solutions that are not only effective but also viable for production.

6.2 Future research direction

In this section, we propose future research directions of robustness from multiple perspectives.

- Expand the training data set: collect test samples from the real world to reduce data bias and distribution bias. Humans are more involved in the model training process, working continuously with the AI model in the feedback process.
- Build a more dynamic, multimodal evaluation system: comprehensively measure the

stability and reliability of the model in different scenarios. The existing static testing methods are difficult to reflect the complexity and uncertainty of the real world, so it is necessary to develop a dynamic evaluation framework that can simulate diversified input conditions and have real-time feedback mechanism.

- Achieving lightweight robust algorithms: optimizing adversarial training and model compression techniques, exploring the uncharted territories in robust optimization algorithms to enhance the model's resistance to noise. Future research could focus on developing efficient small-sample robust training methods, lightweight defense mechanisms that integrate model compression techniques, and more general robust optimization theories.
- Develop AI model system based on human knowledge: it can meet the functions of data input, feature extraction, model training, model reasoning and automatic learning at the same time.
- Continue to explore a more robust basic network structure: improve the tolerance of the model to input disturbances, and enhance its adaptability in the face of unknown attacks or abnormal situations.
- Improving transfer learning algorithms: Solve the problem that AI models trained on a single domain or data set are difficult to generalize and have low robustness, and improve their stable performance in cross-domain tasks.
- Research on deepening model interpretability: reveal the internal relationship between core concepts of trustworthy AI such as robustness, fairness and privacy, and establish a unified theoretical analysis framework. On this basis, develop more intuitive and reliable visualization methods to enhance the understanding of the logic of specific decisions made by the model.
- Robustness from a single model to a complex system level: especially in safety-sensitive fields such as autonomous driving and medical diagnosis, the interaction between each model is taken into account.
- Further exploration of generative AI models: Robust evaluation and enhancement methods for multimodal large models to ensure that their application on the network will not be subjected to malicious attacks and tampering, and ensure the authenticity and security of generated content.

7 Conclusion

Trustworthy AI is crucial for the sustainable development of future AI systems. As deep learning applications expand into security-sensitive areas, ensuring the robustness of AI models has emerged as a pivotal concern for maintaining system reliability. Balancing robustness with other trustworthy AI metrics that can be both complementary and contradictory—remains a significant challenge. This paper we offer a comprehensive review of current robustness issues within AI models, systematically examining recent advancements in assessing and enhancing model robustness from diverse perspectives. Despite rapid progress in AI technologies, robustness challenges persist and require ongoing attention. We delve into robustness evaluation and enhancement methods across various AI architectures, summarizing techniques aimed at improving robustness for models dealing with images, text, and other modalities. These enhancements are discussed through multiple

lenses, including input data collection, data preprocessing, model architecture optimization, adversarial training strategies, and post-model processing. Special emphasis is placed on robustness evaluation and enhancement methods tailored to LLMs due to their growing prominence.Current research trends indicate a shift from 'passive defense' against specific attacks towards developing 'active immunity' by building inherently robust model architectures. Early work primarily focused on defending against known attack vectors; however, emerging approaches now aim to address the root causes of vulnerabilities directly, emphasizing explainability and intrinsic robustness. With interdisciplinary collaboration deepening and AI models becoming increasingly integrated into industry applications, robustness considerations have expanded beyond individual models to encompass system-level robustness across multiple models and software components.Achieving robust AI models capable of meeting the stringent requirements of credible AI, and applicable to safety-critical fields such as autonomous driving and intelligent healthcare—requires continuous theoretical breakthroughs and practical industrial experience accumulation. Only through sustained effort in these areas can we ensure that AI robustness evolves to meet the demands of real-world applications.

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Declarations

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